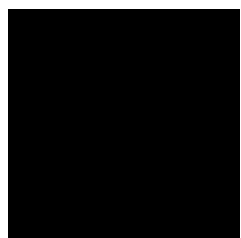


Form H31

(December 2024)



The **ACT**[®]

2024 | 2025

In response to your request for Test Information Release materials, this booklet contains the test questions, scoring keys, and conversion tables used in determining your ACT scores. Enclosed with this booklet is a report that lists each of your answers, shows whether your answer was correct, and, if your answer was not correct, gives the correct answer.

ACT owns the test questions and responses, and you may not share them with anyone in any form.

Directions

This booklet contains tests in English, mathematics, reading, and science. These tests measure skills and abilities highly related to high school course work and success in college. **Calculators may be used on the mathematics test only.**

The questions in each test are numbered, and the suggested answers for each question are lettered. On the answer document, the rows of ovals are numbered to match the questions, and the ovals in each row are lettered to correspond to the suggested answers.

For each question, first decide which answer is best. Next, locate on the answer document the row of ovals numbered the same as the question. Then, locate the oval in that row lettered the same as your answer. Finally, fill in the oval completely. Use a soft lead pencil and make your marks heavy and black. **Do not use ink or a mechanical pencil.**

Mark only one answer to each question. If you change your mind about an answer, erase your first mark thoroughly before marking your new answer. For each question, make certain that you mark in the row of ovals with the same number as the question.

Only responses marked on your answer document will be scored. Your score on each test will be based only on the number of questions you answer correctly during the time allowed for that test. You will **not** be penalized for guessing. **It is to your advantage to answer every question even if you must guess.**

You may work on each test **only** when the testing staff tells you to do so. If you finish a test before time is called for that test, you should use the time remaining to reconsider questions you are uncertain about in that test. You may **not** look back to a test on which time has already been called, and you may **not** go ahead to another test. To do so will disqualify you from the examination.

Lay your pencil down immediately when time is called at the end of each test. You may **not** for any reason fill in or alter ovals for a test after time is called for that test. To do so will disqualify you from the examination.

Do not fold or tear the pages of your test booklet.

**DO NOT OPEN THIS BOOKLET
UNTIL TOLD TO DO SO.**



ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for the underlined part. In most cases, you are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose “NO CHANGE.” In some cases, you will find in the right-hand column a question about the underlined part. You are to choose the best answer to the question.

You will also find questions about a section of the passage, or about the passage as a whole. These questions do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. For many of the questions, you must read several sentences beyond the question to determine the answer. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

Havana’s Camerata Romeu

[1]

As the rhythmic sounds of “Camerata en Guaguancó” ring out in the historic Havana basilica, the visibly thrilled audience listens with rapt attention. [A] It’s December, and Camerata Romeu, an all-woman string chamber orchestra, now performing its annual holiday concert for the

1

city’s thrilled classical music fans.

2

[2]

As a young conductor, Zenaida Romeu wanted to showcase the talent of Cuba’s female musicians as well as that of Cuban and Latin American classical composers. [B] So in 1993, despite the skepticism of critics devoted exclusively to the European classical canon, she founded Camerata Romeu.

1. A. NO CHANGE
B. that is
C. is
D. DELETE the underlined portion.
2. F. NO CHANGE
G. city’s classical music fans, who have assembled in the basilica to hear the orchestra.
H. city of Havana’s classical music fans in the basilica.
J. city’s classical music fans.

[3]

Romeu hoped to bridge the gap she saw
 between classical music and young audiences. Because
 she wanted her musicians to maintain their identities as
 young people themselves, she decided that they would

dress less formally than typical orchestra members.
 Furthermore, they would play all of their pieces from
 memory. Forgoing music stands—Romeu believed,

would rectify the emotional expressiveness of the music
 and foster a more immediate connection with audience
 members of all ages. These approaches worked;
 audiences went to the group's performances.

[4]

Regardless, Camerata Romeu is made up mainly
 of women who join the ensemble while attending the
 prestigious music school at Havana's Higher Institute
 of Art. A spot in Romeu's orchestra is coveted among
 musicians, who must pass a highly selective audition
 process. [C] Playing complex pieces from memory
 absolutely require an extraordinary amount of work
 and dedication. In return, as Romeu mentors
 toward these budding professionals. [D]

3. Given that all the choices are accurate, which one best introduces the main focus of the paragraph?
 - A. NO CHANGE
 - B. was the first woman in Cuba to graduate with an advanced degree in orchestra conducting.
 - C. knew that classical music was not a genre most people associated with Cuba.
 - D. was committed to bringing each composer's vision for a given piece to fruition.
4. F. NO CHANGE
 - G. not be as uptight about what they wore, unlike
 - H. dress down a bit, honestly, compared to
 - J. forget about dressing up as fancily as
5. A. NO CHANGE
 - B. stands, Romeu
 - C. stands Romeu,
 - D. stands Romeu
6. F. NO CHANGE
 - G. enhance
 - H. inflate
 - J. revise
7. Which of the following choices most precisely expresses the idea that audiences were enthusiastic about attending Camerata Romeu's performances?
 - A. NO CHANGE
 - B. assembled at
 - C. turned up at
 - D. flocked to
8. F. NO CHANGE
 - G. For example,
 - H. In contrast,
 - J. DELETE the underlined portion.
9. A. NO CHANGE
 - B. whom
 - C. which
 - D. that
10. F. NO CHANGE
 - G. have always required
 - H. requires
 - J. require
11. A. NO CHANGE
 - B. In return,
 - C. While
 - D. As
12. F. NO CHANGE
 - G. with
 - H. at
 - J. DELETE the underlined portion.

[5]

Over the years, Camerata Romeu has become a cultural institution in Cuba, featuring music created by prominent Latin American composers especially for the orchestra. When it isn't playing at home, the group can be found sharing its special brand of Latin classical music internationally. But the time the orchestra spends, in Havana, culminates in this annual concert,

13

an opportunity for musicians and fans to connect.

14

13. A. NO CHANGE
B. spends in Havana
C. spends in Havana,
D. spends, in Havana

14. F. NO CHANGE
G. this is a great
H. it's an
J. DELETE the underlined portion.

Question 15 asks about the preceding passage as a whole.

15. The writer wants to add the following sentence to the essay:

But the challenge doesn't end there.

The sentence would most logically be placed at:

- A. Point A in Paragraph 1.
B. Point B in Paragraph 2.
C. Point C in Paragraph 4.
D. Point D in Paragraph 4.

PASSAGE II**Nest in the Sand**

Driving slowly along an empty beach on Kiawah Island, South Carolina, I notice an area of scattered sand among the dunes: a sign of a potential sea turtle nesting site. As a volunteer conservationist working to protect endangered loggerhead sea turtles, I patrol the beach before sunrise, scanning for evidence of any nests created overnight. It rained yesterday, so any marks I see this morning were surely made last night.

The area of disturbed sand is practically forty yards
¹⁶
 from the water's edge. It's not uncommon for a female
 loggerhead, weighing up to three hundred pounds, to

make a slow and exhausting crawl this far onto the beach
¹⁷
 to find a safe nesting spot. Once she selects a site, the

turtle uses her flippers to dig a large pit, she lays a clutch
¹⁸
 of nearly one hundred eggs. To conceal the site, she

throws sand all around the area. 19

I approach the area on foot, cautiously. Though the
 extent of sand thrown about are characteristics of a turtle's
²⁰
 dig site, I can't be sure I've found a nest yet. Sometimes

sea turtles create what's called: a "false crawl." For
²¹
 one reason or another, a loggerhead may plod onto the

beach 22 without laying eggs. This site is promising,

equally, because of something I notice about the tracks
²³

16. Which choice makes most clear that the narrator thinks the area of disturbed sand may be more than forty yards from the water's edge?

- F. NO CHANGE
- G. close to
- H. at least
- J. nearly

17. The writer is considering deleting the underlined portion. Should the underlined portion be kept or deleted?

- A. Kept, because it indicates the length of time it generally takes for a sea turtle to pick a nesting spot.
- B. Kept, because it helps emphasize how arduous the journey to find a nesting spot is for a sea turtle.
- C. Deleted, because it detracts from the paragraph's focus on the process a sea turtle uses to find a nesting spot.
- D. Deleted, because it implies that some sea turtles choose their nesting spots close to the water's edge.

18. F. NO CHANGE

- G. pit into which she
- H. pit, which she
- J. pit,

19. Given that all the following choices are accurate, which one, if added here, would best conclude the paragraph by indicating the final actions taken by the nesting sea turtle?

- A. It may look as though she's crying, but her "tears" are flushing away excess seawater and sand.
- B. Female loggerhead turtles typically nest on the island between May and October each year.
- C. She then trudges back to sea, never to return to the nest.
- D. Later on, the eggs will hatch.

20. F. NO CHANGE

- G. are characteristic
- H. is characteristics
- J. is characteristic

21. A. NO CHANGE

- B. what is called,
- C. what's called,
- D. what's called

22. If the writer were to add the phrase "and even dig a hole" at this point, the paragraph would primarily gain:

- F. information about how the indications of a loggerhead nesting site can be misleading.
- G. a detail that helps emphasize how rare it is for the narrator to locate a loggerhead nest.
- H. a reminder that the narrator has spotted a nest but has yet to locate the mother turtle.
- J. an indication of why loggerheads dig holes for purposes other than nesting.

23. A. NO CHANGE

- B. unfortunately,
- C. furthermore,
- D. though,

that mark the turtle's approach and their withdrawal.

The tracks leading to the site are marked by deep,
comma-shaped divots in the sand, evidence of the turtle
hoisting her belly up so as not to drag her eggs on the
ground. The return tracks, in other words, reveal that
the turtle dragged herself back to sea. The shallow path

in the sand that was made smooth by her prostrate belly,
is a clear sign she's left eggs behind.

[1] If left undisturbed, the turtles will begin to hatch
in two months. [2] Having found the well-hidden nest, I
record it's location, the first step in keeping the eggs safe.
[3] They'll emerge from underground as a group and

follow the light of the moon to the sea. 29

24. F. NO CHANGE
G. this
H. it's
J. DELETE the underlined portion.

25. A. NO CHANGE
B. consequently,
C. in contrast,
D. therefore,

26. F. NO CHANGE
G. sand that,
H. sand,
J. sand

27. A. NO CHANGE
B. turtle's will begin to hatch
C. turtles will begin to hatch,
D. turtles' will begin to hatch,

28. F. NO CHANGE
G. where its' located,
H. where its located,
J. its location,

29. Which sequence of sentences makes this paragraph most logical?
A. NO CHANGE
B. 2, 1, 3
C. 2, 3, 1
D. 3, 2, 1

Question 30 asks about the preceding passage as a whole.

30. Suppose the writer's primary purpose had been to portray someone working to protect an endangered species. Would this essay accomplish that purpose?
F. Yes, because it clarifies how to distinguish between the tracks and nests of endangered loggerhead turtles and those of other sea turtles.
G. Yes, because it provides a firsthand account of a volunteer's efforts to locate the nests of endangered loggerhead turtles.
H. No, because it primarily focuses on explaining the positive impact of efforts to protect loggerhead turtle nests.
J. No, because it primarily focuses on describing the reasons loggerhead sea turtles are endangered.

PASSAGE III

A Cut Above the Rest

In the first half of the nineteenth century, as silhouette³¹ portraits grew popular in the United States. With just paper and scissors, a skilled silhouettist could craft a convincing outline of a person in minutes. Although the art could be³² produced quickly with simple materials, it cost little. A large portion of the populace could afford silhouettes of themselves or of loved ones.

Perhaps the most remarkable silhouettist of the era was William Henry Brown, born in South Carolina in 1808. As a teenager, Brown displayed an impressive³³

ability to produce silhouettes, as he put it, “with³⁴ wonderful facility and accuracy.” He turned his

talent into a career. 35 A key feature of Brown’s

silhouettes³⁶ were that instead of depicting only a

subject’s head and torso³⁷, the usual style—Brown cut full-length portraits. Unlike the static profiles seen in

31. A. NO CHANGE
B. The first half of the nineteenth century, in which
C. When, in the first half of the nineteenth century,
D. In the first half of the nineteenth century,
32. F. NO CHANGE
G. Because
H. While
J. As if
33. Which choice most clearly emphasizes that Brown’s talent came naturally to him?
A. NO CHANGE
B. an instinctive
C. an unusual
D. a refined
34. Which of the following alternatives to the underlined portion would NOT be acceptable?
F. he could make them, he said,
G. claiming he could make them
H. making them, in his words,
J. he claimed,
35. At this point, the writer is considering dividing the paragraph into two. Should the writer begin or not begin a new paragraph here?
A. Begin a new paragraph, because the essay shifts here from an outline of Brown’s early career to an evaluation of how the style of his silhouettes evolved over the course of his career.
B. Begin a new paragraph, because the essay shifts here from information about Brown’s life and the start of his vocation to a consideration of the characteristics of his silhouettes.
C. DO NOT begin a new paragraph, because doing so would interrupt the discussion of the ways Brown’s upbringing influenced the subject matter he chose for his silhouettes.
D. DO NOT begin a new paragraph, because doing so would interrupt the account of how Brown learned the art of making silhouettes.
36. F. NO CHANGE
G. have been
H. was
J. are
37. A. NO CHANGE
B. head and torso—
C. head, and torso,
D. head, and torso

most silhouettes, Brown's figures stood, sat, walked, and rode horses. In particular, his 1846 book, *Portrait Gallery of Distinguished American Citizens*, displayed the potential of the full-length format. An astute observer, Brown captured each subject's character through details³⁸ of posture, dress, gesture, and movement.

Brown didn't limit himself to depicting lone individuals either. His twenty-five-foot-long portrayal of the St. Louis Fire Company, including a fire engine³⁹

and sixty-five men. Additionally, Brown supplemented⁴⁰ some of his silhouettes with paint and ink, such as in *Hauling the Whole Week's Picking*, which showed nine people, four mules, three dogs, two oxen, and two wagons containing⁴¹ cotton. The paint, in a range of brown and

yellow tones, bring a breath of immediacy⁴² into the scene, as if the workers have just left the field.

In the 1850s, photography became inexpensive, and demand for silhouettes nearly disappeared. Fortunately, Brown found work as a train engineer, a profession he had trained for as a youth. Even more, fortunately⁴³ he had

already created a body of work, his silhouettes represent⁴⁴ the pinnacle of the art.

38. F. NO CHANGE
 G. Brown's style captured the character of each subject
 H. the silhouettes captured each subject's character
 J. each subject's character was captured
39. A. NO CHANGE
 B. Company, which includes
 C. Company that included
 D. Company included
40. F. NO CHANGE
 G. as in *Hauling the Whole Week's Picking*, Brown supplemented
 H. with paint and ink, Brown supplemented
 J. Brown supplemented, furthermore,
41. Which choice most clearly conveys that the wagons carried substantial loads of cotton?
 A. NO CHANGE
 B. with cargos of
 C. burdened with
 D. conveying
42. F. NO CHANGE
 G. introduce a breath of
 H. breathes
 J. breathe
43. A. NO CHANGE
 B. Even, more fortunately,
 C. Even, more fortunately
 D. Even more fortunately,
44. F. NO CHANGE
 G. work, such is a representation of
 H. work, these images represent
 J. work that represents

Question 45 asks about the preceding passage as a whole.

45. Suppose the writer's primary purpose had been to detail the history of a particular art form. Would this essay accomplish that purpose?
- A. Yes, because it describes the era in which silhouette-making was most popular and some of the most famous silhouettes that were produced at the time.
 - B. Yes, because it follows the history of silhouette-making from its invention to its decline.
 - C. No, because while it provides some historical information about silhouette-making, it mainly focuses on the work of one exceptional silhouettist.
 - D. No, because while it describes some important historical silhouettes, it mainly focuses on the process of creating silhouettes.

PASSAGE IV

Swimming with the Sharks

[1] Instead of recognizing sharks as complex creatures worthy of respect and study, people largely considered them primitive brutes. [2] Before the 1950s, misinformation about sharks were pervasive.⁴⁶ [3] Dr. Eugenie Clark, a pioneer in marine science, dedicated her career to changing this widespread misperception of the animals. 47

Clark's fascination with sharks was ignited in childhood; during weekly visits to the New York Aquarium, she spent hours watching the animals.

46. F. NO CHANGE
G. toward sharks was
H. about sharks was
J. on sharks were
47. Which sequence of sentences makes this paragraph most logical?
- A. NO CHANGE
 - B. 2, 1, 3
 - C. 2, 3, 1
 - D. 3, 2, 1

Of Japanese heritage, Clark credited her interest in studying marine biology to a key facet of Japanese
48 culture, earning her doctorate in 1950. A passionate and

inquisitive researcher, Clark once founded the Cape Haze
49 Marine Laboratory in Florida. There, her research on sharks led to groundbreaking discoveries.

In 1958, for instance, Clark helmed an experiment in which two lemon sharks were trained to press an underwater target to obtain food. She later expanded the study, teaching sharks to visually
50

distinguish between targets of varying shapes, and colors.
51

These experiments have been repeated by many other
52 scientists over the years.
52

Using these findings about shark intelligence as a springboard: Clark began
53

publicly advocating for the misunderstood animal.
54

48. The writer is considering revising the underlined portion to the following:

the sea's importance in

Given that the information creates a sentence that is still accurate, should the writer make this revision?

- F. Yes, because the revision helps clarify the connection between Clark's interest in marine biology and her Japanese heritage.
- G. Yes, because the revision explains why the sea plays an important role in Japanese culture.
- H. No, because the revision detracts from the discussion of Clark's interest in marine biology.
- J. No, because the revision conflicts with the essay's indication that Clark was interested in the sea since childhood.

49. Which choice most clearly indicates that Clark founded the laboratory shortly after earning her doctorate?

- A. NO CHANGE
- B. eventually
- C. recently
- D. soon

50. F. NO CHANGE
G. this time, she taught
H. she was teaching
J. she taught

51. A. NO CHANGE
B. distinguish—between targets of varying shapes
C. distinguish between targets, of varying shapes,
D. distinguish between targets of varying shapes

52. Given that all the choices are accurate, which one provides the best transition between this paragraph and the next paragraph?

- F. NO CHANGE
- G. undisputedly refuted the long-held myth that sharks were incapable of higher-order learning.
- H. clearly demonstrated that further studies of shark intelligence were warranted.
- J. were later conducted upon nurse sharks and sandbar sharks.

53. A. NO CHANGE
B. springboard;
C. springboard,
D. springboard

54. The writer wants to emphasize that Clark began working actively on behalf of sharks in broadly visible ways. Which choice best accomplishes that goal?

- F. NO CHANGE
- G. relating her professional views on
- H. telling people about
- J. defending

Whereas her message, shared through popular books,
55

articles, and lectures, gradually took effect, sharks'
56
ill-deserved reputation began to fade. Meanwhile, the prominent scientist continued her trailblazing research, advancing a variety of scientific knowledge regarding shark behavior and biology.

Clark's legacy lies not only with the animals' whose cause she championed
57

also with the generations of scientists she inspired.
58

Sighting her lifelong love of the ocean as motivation,
59

Clark continued teaching and scuba diving into her
60
nineties. With her trademark enthusiasm, she encouraged people to enjoy and not fear the sea and its inhabitants: "When you see a shark underwater, you should say, 'How lucky I am to see this beautiful animal!'"

PASSAGE V

The following paragraphs may or may not be in the most logical order. Each paragraph is numbered in brackets, and question 75 will ask you to choose the sequence of paragraphs that makes the essay most logical.

The Magic of the Golden Hour

[1]

When working outside, photographers are at the mercy of natural light. Weather, season, and time of day all affect the light, because serious photographers know
61
that during the so-called "golden hour" the light is likely

55. A. NO CHANGE
B. Although
C. Until
D. As

56. F. NO CHANGE
G. had an affect,
H. took affect,
J. effected,

57. A. NO CHANGE
B. animals whom's
C. animals whose
D. animals who's

58. F. NO CHANGE
G. additionally
H. as well as
J. but also

59. A. NO CHANGE
B. Having sited
C. Citing
D. Siting

60. F. NO CHANGE
G. teaching and scuba diving remained interests of Clark's
H. Clark's teaching and scuba diving continued
J. teaching and scuba diving were activities Clark continued

61. A. NO CHANGE
B. but
C. for
D. if

to be photo friendly. The golden hour comes twice a day—just after sunrise and just before sunset—and features soft ⁶² yellow-and-orange light that makes for luminous and dramatic photographs.

[2]

The light of the golden hour is unique because of variations in how long the sun's photons—particles of light—spends traversing Earth's atmosphere. When the sun shines high overhead, the photons pass nearly straight ⁶³ down through the atmosphere. When the sun shines near the horizon, nonetheless, the photons travel at a low ⁶⁴

angle relative, to the ground, increasing the amount of ⁶⁵ time they're in the atmosphere.

[3]

When photographers look at pictures taken during the golden hour, they outright ignore blue-light ⁶⁶ scattering and diffusion. They just relish the results,

the magical effects of the golden hour. Honey-tinted ⁶⁷ light lends skin a glowing smoothness, soft contrast infuses scenes with nuance and emotion, and

long shadows that conjure a sense of depth. ⁶⁸

[4]

The increased time the photons take to pass ⁶⁹ through the atmosphere, the more likely they are to

smash right into atoms and molecules. Such collisions ⁷⁰

62. F. NO CHANGE
G. pinpoints
H. promotes
J. endorses

63. A. NO CHANGE
B. has spent
C. spend
D. is

64. F. NO CHANGE
G. on the other hand,
H. in any event,
J. especially,

65. A. NO CHANGE
B. angle relative to the ground,
C. angle, relative to the ground
D. angle relative to the ground

66. Which choice most clearly indicates this sentence is speculative rather than factual?

- F. NO CHANGE
G. probably don't think about
H. definitely don't ponder
J. disregard

67. Which of the following alternatives to the underlined portion would NOT be acceptable?

- A. magical effects produced by the golden hour.
B. magical effects the golden hour creates.
C. golden hour generates magical effects.
D. golden hour's magical effects.

68. F. NO CHANGE
G. the conjuring, by long shadows, of a sense of depth.
H. a sense of depth conjured by long shadows.
J. long shadows conjure a sense of depth.

69. A. NO CHANGE
B. length of
C. more
D. DELETE the underlined portion.

70. F. NO CHANGE
G. knock heads with
H. strike
J. bonk



have two effects. The scattering of violet and blue light produce the most obvious effect. Because these colors

⁷¹

of photons are more susceptible to being scattered by

atmospheric collisions, a significant portion of them

⁷²

don't reach the ground during the golden hour. 73

That leaves yellow, orange, and red as the dominant

colors at ground level.

⁷⁴

[5]

The second effect of the collisions is that many photons get deflected just slightly from their straight courses, arriving at the ground via many different angles. This deflected light is called “diffuse” light, and it’s why shadows during the golden hour have soft edges and gently lit interiors.

71. A. NO CHANGE
B. represent
C. are
D. is
72. F. NO CHANGE
G. collisions, resulting in
H. collisions, meaning
J. collisions that
73. At this point, the writer is considering adding the following sentence:

Some photographers go to great lengths to supplement natural light with flashes, reflectors, and other tools.

Given that the information is accurate, should the writer make this addition here?

A. Yes, because it illustrates how photographers counteract the scattering of violet and blue photons.
B. Yes, because it provides further evidence of the trouble that natural light can cause photographers.
C. No, because it disrupts the paragraph’s discussion of the causes and consequences of the scattering of violet and blue photons.
D. No, because it shows that photographers have more control over natural light than the paragraph previously stated.
74. F. NO CHANGE
G. at ground level, especially yellow, orange, and red.
H. at ground level, more prevalent than other colors.
J. left at ground level.

Question 75 asks about the preceding passage as a whole.

75. For the sake of logic and cohesion, Paragraph 3 should be placed:
A. before Paragraph 1.
B. after Paragraph 1.
C. after Paragraph 4.
D. after Paragraph 5.

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.



MATHEMATICS TEST

60 Minutes—60 Questions

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. The function f is defined below.

$$f(x) = |x|$$

What is(are) the output value(s) of $f(2)$?

- A. $\frac{1}{2}$
- B. 2
- C. $|f|(2)$
- D. $\frac{|2|}{f}$
- E. $\frac{|2|}{x}$

DO YOUR FIGURING HERE.

2. Quinn bought a tractor. He made an initial payment of \$900. He then made 36 equal monthly payments. Quinn paid a total amount of \$9,000 for the tractor. What was the amount of each of his monthly payments?

- F. \$ 25
- G. \$225
- H. \$250
- J. \$275
- K. \$360

3. A new operation, $\diamond$, is defined for 2 ordered pairs of integers as follows: $(a,b) \diamond (c,d) = \frac{ac+bd}{ab-cd}$. What is the value of $(4,3) \diamond (1,2)$?

- A. -5
- B. -2
- C. 1
- D. 2
- E. 10



DO YOUR FIGURING HERE.

4. To determine a student's overall test score for the semester, Mr. Weintraub omits the lowest test score and calculates the average of the remaining test scores. Holly took all 5 tests and earned the following test scores in Mr. Weintraub's class this semester: 71, 76, 81, 84, and 95. What overall test score did Holly earn in Mr. Weintraub's class this semester?

F. 81.0
G. 81.4
H. 83.0
J. 84.0
K. 85.5

5. A rectangular field measures 300 feet by 200 feet. A rectangular portion of the field measuring 225 feet by 175 feet is paved for a parking lot, and the rest of the field is left unpaved. How many square feet of the field are left unpaved?

A. 16,000
B. 20,625
C. 30,625
D. 32,500
E. 39,375

6. All 26 students in a high school music department are enrolled in choir, band, or both. Of the 26 students, 14 are enrolled in choir and 18 are enrolled in band. How many are enrolled in both choir and band?

F. 4
G. 6
H. 14
J. 20
K. 26

7. Jesse has 4 shirts, 8 pairs of jeans, and 4 coats, which all go together well. How many different outfits consisting of 1 of his 4 shirts, 1 of his 8 pairs of jeans, and 1 of his 4 coats are available for Jesse to wear?

A. 12
B. 16
C. 32
D. 64
E. 128

8. What is the product of $(3x + 2)$ and $(3x - 2)$?

F. $6x^2 - 4$
G. $9x^2 - 4$
H. $9x^2 + 4$
J. $9x^2 - 12x - 4$
K. $9x^2 + 12x - 4$

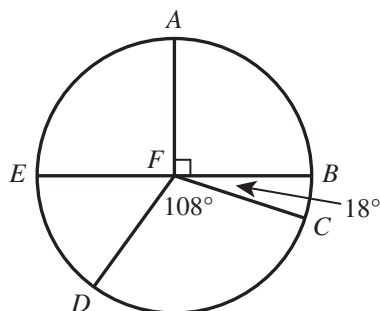


9. Zhu has \$25 to spend on camping supplies. The table below shows the prices, including tax, of some items. Given that Zhu buys 1 hand broom, 1 rubber mallet, and 2 telescoping forks, what is the greatest number of tent stakes she can buy with her remaining money?

Item	Price
Hand broom	\$3.45
Rubber mallet	\$6.00
Telescoping fork	\$3.25
Tent stake	\$1.20

- A. 7
B. 8
C. 9
D. 10
E. 11
10. There were 425 students in the 10th grade at Hayes County High School this year. Of these students, $\frac{3}{5}$ were enrolled in geometry. Of the students enrolled in geometry, $\frac{1}{5}$ earned a grade of A in geometry this year. How many of the geometry students earned a grade of A in geometry this year?
- F. 51
G. 85
H. 136
J. 170
K. 340

11. A circle centered at F with a diameter of $\overline{EB}$ is shown below. The measure of $\angle BFC$ is 18° , and the measure of $\angle CFD$ is 108° . Angle $\angle AFB$ is a right angle. What is the measure, in degrees, of $\widehat{DEA}$?



- A. 108°
B. 144°
C. 160°
D. 216°
E. 288°

DO YOUR FIGURING HERE.



12. Which of the following expressions gives the number of revolutions a tire with a diameter of d feet will make in a trip m miles long?

(Note: 1 mile = 5,280 feet)

F. $\frac{5,280m}{\pi d}$

G. $\frac{5,280d}{\pi m}$

H. $\frac{5,280}{\pi dm}$

J. $\frac{\pi m}{5,280d}$

K. $\frac{\pi d}{5,280m}$

DO YOUR FIGURING HERE.

13. Which of the following matrices is equal to $3\begin{bmatrix} -2 & 1 \\ 0 & -5 \end{bmatrix}$?

A. $\begin{bmatrix} -6 & -12 \end{bmatrix}$

B. $\begin{bmatrix} -3 \\ -15 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 4 \\ 3 & -2 \end{bmatrix}$

D. $\begin{bmatrix} -\frac{2}{3} & \frac{1}{3} \\ 0 & -\frac{5}{3} \end{bmatrix}$

E. $\begin{bmatrix} -6 & 3 \\ 0 & -15 \end{bmatrix}$

14. A vertical rod 42 feet tall casts a shadow 14 feet long, and at the same time a nearby vertical flagpole casts a shadow 42 feet long. How many feet tall is the flagpole?

F. 14

G. 28

H. 42

J. 70

K. 126

15. Which of the following (x,y) pairs is the solution for the system of equations $x + 2y = 8$ and $-2x + y = 14$?

A. $(-4,6)$

B. $(-1,4.5)$

C. $(1,3.5)$

D. $(6,1)$

E. $(8,0)$

16. Which of the following expressions is equivalent to $x^2 - 13x + 30$?

F. $(x - 15)(x - 2)$

G. $(x - 15)(x + 2)$

H. $(x - 10)(x - 3)$

J. $(x - 10)(x + 3)$

K. $(x + 10)(x - 3)$



17. A circle with the equation $x^2 + y^2 = 36$ is graphed in the standard (x,y) coordinate plane. At what points does the circle intersect the x -axis?

A. $(-3,0)$ and $(3,0)$
 B. $(-6,0)$ and $(6,0)$
 C. $(-12,0)$ and $(12,0)$
 D. $(-18,0)$ and $(18,0)$
 E. $(-36,0)$ and $(36,0)$

DO YOUR FIGURING HERE.

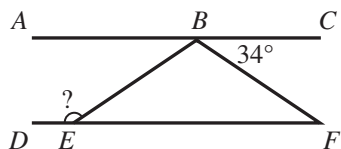
18. A certain company manufactured 1,000 vacuums and recorded the number of those vacuums sold within 5 months. The company's findings are summarized in the table below. What is the probability 1 of the 1,000 vacuums was sold 3 or 4 months after it was manufactured?

Month	No. of vacuums sold
1	56
2	173
3	301
4	297
5	103

F. 0.297
 G. 0.301
 H. 0.598
 J. 0.701
 K. 0.827

19. In the figure below, B lies on $\overline{AC}$, E lies on $\overline{DF}$, $\overline{AC} \parallel \overline{DF}$, $\triangle EBF$ is isosceles with $\overline{BE} \cong \overline{BF}$, and $\angle CBF$ measures 34° . What is the measure of $\angle BED$?

A. 107°
 B. 112°
 C. 124°
 D. 134°
 E. 146°



20. Let y be an irrational number. Which of the following statements about the value of $5y$ is true?

F. It is rational.
 G. It is irrational.
 H. It is imaginary.
 J. It is undefined.
 K. It is neither irrational nor rational.

21. A certain bus traveled at an average speed of 45 miles per hour. What was the average speed, in feet per second, traveled by the bus?

(Note: 1 mile = 5,280 feet)

A. 30
 B. 45
 C. 66
 D. 90
 E. 120



22. Given that $h(z) = (-5z) + 8$ and $g(z) = -2z^2 + 2z + 8$, what is $(h + g)(z)$?

F. $-5z + 16$
 G. $-7z^2 + 2z + 16$
 H. $-2z^2 - 3z + 16$
 J. $-5z^3 + 16$
 K. $-5z^4 + 16$

DO YOUR FIGURING HERE.

23. A theme park is building 3 new rides: the Thunder, the Gateway, and the Cyclops. Guests to the park were asked to respond “yes,” “no,” or “maybe” when asked if they would like to experience one of these rides. The results are shown in the table below. To the nearest 0.01, what is the probability that a randomly selected guest who was asked about the Gateway responded “yes”?

	Thunder	Gateway	Cyclops	Total
Yes	52	98	101	251
No	55	43	21	119
Maybe	55	28	49	132
Total	162	169	171	502

- A. 0.20
 B. 0.39
 C. 0.58
 D. 0.64
 E. 0.98
24. A certain right circular cone has a volume of 616 cubic meters. Given that the cone has a height of 5 meters, which of the following gives the radius of the base of the cone, in meters?

(Note: The volume of a right circular cone with radius r and height h is $\frac{1}{3}\pi r^2 h$.)

F. $\sqrt{\frac{616}{5}}$
 G. $\sqrt{\frac{616(3)}{5\pi}}$
 H. $\frac{1}{2}\left(\frac{616}{5\pi}\right)$
 J. $2\sqrt{\frac{616(3)}{5\pi}}$
 K. $\pi\sqrt{\frac{616(3)}{5\pi}}$

25. $3[\sin^2(2\theta) + \cos^2(2\theta)] = ?$

A. 0
 B. 1
 C. 2
 D. 3
 E. 6



26. In the equation $w = 6m + 9$, w represents the total weight, in ounces, of a carton of cookies where each cookie weighs the same amount, and m represents the number of cookies in the carton. Which of the following quantities is represented by 9 in the equation?

F. The number of cartons
 G. The number of cookies in the carton
 H. The weight, in ounces, of 1 cookie
 J. The weight, in ounces, of 9 cookies
 K. The weight, in ounces, of the empty carton

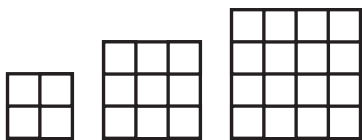
DO YOUR FIGURING HERE.

27. For an angle with measure α in a right triangle,

$\sin \alpha = \frac{144}{145}$ and $\tan \alpha = \frac{144}{17}$. What is the value of $\cos \alpha$?

A. $\frac{17}{145}$
 B. $\frac{17}{144}$
 C. $\frac{17}{\sqrt{41,761}}$
 D. $\frac{17}{\sqrt{20,447}}$
 E. $\frac{145}{17}$

28. The first 3 elements of a square grid pattern are shown in order from left to right below, where a 1×1 square is the smallest square shown in any grid.



What is the total number of 1×1 squares in the first 6 elements?

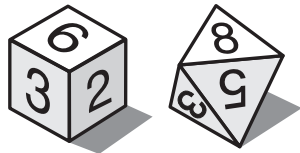
F. 68
 G. 71
 H. 91
 J. 139
 K. 140

29. The length of Rectangle A is equal to the length of Rectangle B. The width of Rectangle A is 6 meters less than the width of Rectangle B. If it can be determined, how many meters less is the perimeter of Rectangle A than the perimeter of Rectangle B?

A. 6
 B. 12
 C. 24
 D. 36
 E. Cannot be determined from the given information

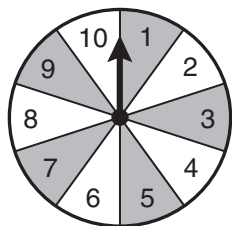


30. Shown below are 2 fair dice. One has 6 faces, each numbered with a different integer from 1 through 6. The other has 8 faces, each numbered with a different integer from 1 through 8. What is the probability that, when the 2 dice are rolled, the sum of the faceup integers will be 12 ?



DO YOUR FIGURING HERE.

- F. $\frac{1}{48}$
 G. $\frac{2}{48}$
 H. $\frac{3}{48}$
 J. $\frac{5}{48}$
 K. $\frac{9}{48}$
31. A spinner has 10 congruent sectors that have alternating shading and are numbered 1 through 10, as shown below. The pointer is spun. Given that the pointer stops in a shaded sector, what is the probability that the pointer stops in a sector that is a multiple of 3 ?

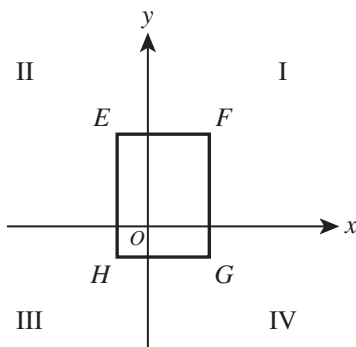


- A. $\frac{1}{10}$
 B. $\frac{2}{3}$
 C. $\frac{2}{5}$
 D. $\frac{2}{10}$
 E. $\frac{3}{20}$
32. Given $f(x) = x + 2$ and $g(x) = \frac{1}{x^2}$, where $x > 0$, what is $g(f(x))$?

- F. $x^2(x + 2)$
 G. $\frac{1}{x^2} + 2$
 H. $\frac{1}{(x + 2)^2}$
 J. $\frac{(x + 2)}{x^2}$
 K. $\frac{1 + x^2(x + 2)}{x^2}$



33. Rectangle $EFGH$ with vertices at $(-1,3)$, $(2,3)$, $(2,-1)$, and $(-1,-1)$ is shown in the standard (x,y) coordinate plane below. The quadrants of the coordinate plane are labeled. What percent of the rectangle's area lies in Quadrant IV?



DO YOUR FIGURING HERE.

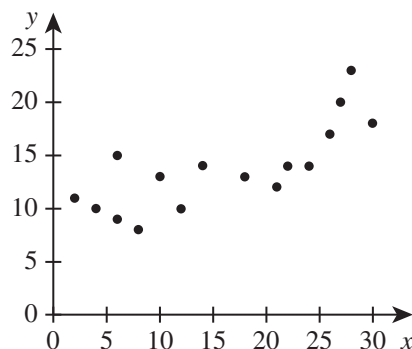
- A. $8\frac{1}{3}\%$
 B. $11\frac{1}{9}\%$
 C. $16\frac{2}{3}\%$
 D. $22\frac{2}{9}\%$
 E. 25%

34. Which of the following expressions is equivalent to

$$\frac{4}{c^2} \left(\frac{c+2}{c} \right) ?$$

- F. $\frac{8}{c^2}$
 G. $\frac{4c^2+8c}{c^2}$
 H. $\frac{4c+2}{c^3}$
 J. $\frac{4c+8}{c^2}$
 K. $\frac{4c+8}{c^3}$

35. Which of the following values is closest to the slope of the line of best fit (with the least sum of squared residuals) for the data represented by the scatterplot in the standard (x,y) coordinate plane below?



- A. $-\frac{1}{4}$
 B. $-\frac{1}{2}$
 C. 0
 D. $\frac{1}{3}$
 E. 1
36. Given $i^2 = -1$, which of the following expressions is equivalent to $(9i - 2)(3 - 5i)$?

- F. $37i$
 G. $4i + 1$
 H. $37i - 51$
 J. $37i + 39$
 K. $-45i - 6$



37. For what value of x , if any, is the function below undefined?

$$f(x) = \frac{(-1)(x+2)}{(x+2)}$$

- A. -2
- B. -1
- C. 1
- D. 2
- E. f is defined for all real numbers.

DO YOUR FIGURING HERE.

38. The Chocolate Mousse Lodge sells gift boxes of chocolates. The actual cardboard box is free, and there is a fixed cost for each gourmet chocolate and another fixed cost for each premiere chocolate in each gift box. A gift box with 5 gourmet chocolates and 10 premiere chocolates costs \$30.35, and a gift box with 6 gourmet chocolates and 6 premiere chocolates costs \$22.68. What is the cost of a gift box with 2 gourmet chocolates and 1 premiere chocolate?

- F. \$1.53
- G. \$2.56
- H. \$4.60
- J. \$5.27
- K. \$5.80

39. Which of the following expressions is NOT equivalent

to $\left(\frac{4^9}{4^{15}}\right)^{\frac{1}{3}}$?

- A. $(4^{-6})^{\frac{1}{3}}$
- B. 4^{-2}
- C. $\frac{1}{4^2}$
- D. $\frac{4^3}{4^5}$
- E. $(4^{24})^{\frac{1}{3}}$

40. Given the relatively prime positive integers a , b , c , and d such that $a > b > c > d$, which of the following fractions has the greatest value?

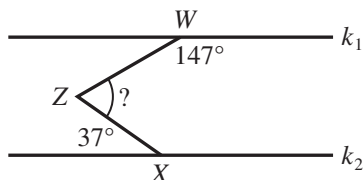
- F. $\frac{a}{b}$
- G. $\frac{a}{c}$
- H. $\frac{b}{c}$
- J. $\frac{c}{a}$
- K. $\frac{d}{a}$



41. In the figure below, k_1 and k_2 are parallel lines, and W and X are points on k_1 and k_2 , respectively. In the same plane, point Z lies between k_1 and k_2 , and some angle measures are given. What is the degree measure of $\angle XZW$?

(Note: The figure is NOT drawn to scale.)

- A. 53°
 B. 60°
 C. 70°
 D. 92°
 E. 110°

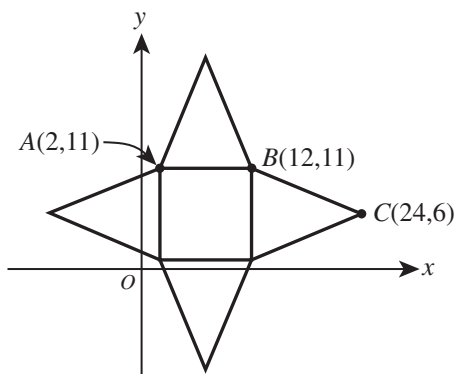


42. What are all the values of z that make the equation below true?

$$z^2 + z + 3 = 0$$

- F. -3 and 0
 G. -2 and -1
 H. 1 and 2
 J. $-2i$ and $2i$
 K. $\frac{-1 - i\sqrt{11}}{2}$ and $\frac{-1 + i\sqrt{11}}{2}$

43. A net of a square pyramid with 4 congruent triangular sides is shown below in the standard (x,y) coordinate plane. Points A , B , and C are vertices of the net. What is the total surface area, in square coordinate units, of the pyramid?



44. Let function f be defined by $f(x) = 4x + 3$. Let function g be defined by $g(x) = -4x^2 + 10x + 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at one of the following (x,y) points. Which one?

- F. $(1, 7)$
 G. $(7, 1)$
 H. $(1, \frac{1}{2})$
 J. $(\frac{1}{2}, 1)$
 K. $(\frac{1}{2}, -1)$

DO YOUR FIGURING HERE.

GO ON TO THE NEXT PAGE.



45. Each of the following sets contains the side lengths, in inches, of a right triangle EXCEPT one set. Which one?

- A. $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}\right\}$
- B. $\left\{\frac{3}{7}, \frac{4}{7}, \frac{5}{7}\right\}$
- C. $\left\{\frac{5}{12}, 1, \frac{13}{12}\right\}$
- D. $\left\{\frac{6}{11}, \frac{8}{11}, \frac{10}{11}\right\}$
- E. $\left\{2, \frac{15}{4}, \frac{17}{4}\right\}$

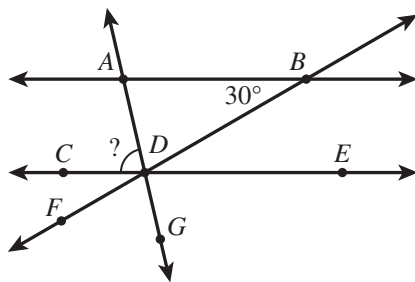
46. Consider the function f defined by $f(x) = -\frac{1}{3}x - 5$ with domain $-6 \leq x \leq 9$. What is the range of f ?

- F. $-42 \leq f(x) \leq 3$
- G. $-8 \leq f(x) \leq -3$
- H. $-6 \leq f(x) \leq -3$
- J. $-6 \leq f(x) \leq 3$
- K. $-6 \leq f(x) \leq 9$

47. For $\sqrt{-1} = i$, and for certain real numbers a and b , the equation $(10 - i) + (8 + 6i) = a + bi$ is true. What is the value of ab ?

- A. 86
- B. 90
- C. 108
- D. 112
- E. 126

48. In the figure below, $\overleftrightarrow{AB}$ is parallel to $\overleftrightarrow{CE}$; $m\angle ABF = 30^\circ$; and lines $\overleftrightarrow{AG}$, $\overleftrightarrow{BF}$, and $\overleftrightarrow{CE}$ intersect at D . If it can be determined, what is $m\angle ADC$?



- F. 30°
- G. 60°
- H. 75°
- J. 150°
- K. Cannot be determined from the given information

49. Which of the following values is equal to the remainder when $x^7 - x^6 - 2x^4 + 2x^3 + x - 1$ is divided by $x - 1$?

- A. -8
- B. -1
- C. 0
- D. 2
- E. 8

DO YOUR FIGURING HERE.



50. For a certain normally distributed data set, the mean is 100, and the standard deviation is 20. Given that 68% of the data are within 1 standard deviation of the mean, 68% of the data are within which of the following intervals?

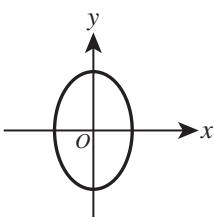
F. (0, 68)
 G. (68, 132)
 H. (80, 100)
 J. (80, 120)
 K. (100, 120)

DO YOUR FIGURING HERE.

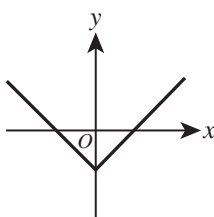
51. One of the following is a hyperbola graphed in the standard (x,y) coordinate plane. Which one?

(Note: All key features of the graphs are shown.)

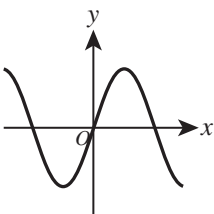
A.



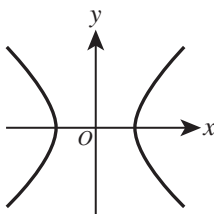
D.



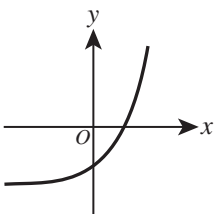
B.



E.



C.



52. Let a , b , c , and d be real numbers greater than or equal to 2. Four functions are listed below.

$$f(x) = ab^x$$

$$g(x) = ax + b$$

$$h(x) = ax^2 + bx + c$$

$$k(x) = ax^3 + bx^2 + cx + d$$

The values of which of the listed functions will exceed the values of all the other listed functions as x approaches infinity?

F. $f(x) = ab^x$
 G. $g(x) = ax + b$
 H. $h(x) = ax^2 + bx + c$
 J. $k(x) = ax^3 + bx^2 + cx + d$
 K. Cannot be determined from the given information



Use the following information to answer questions 53–55.

DO YOUR FIGURING HERE.

Shakir and Julia have completed a typing class. The table below gives their typing speeds in words per minute (wpm) on different days during the 99-day class.

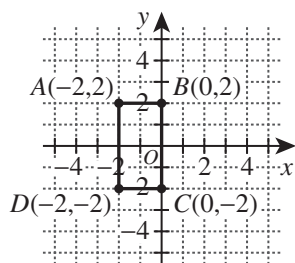
(Note: The typing speed of each student in the class was measured at the end of each of the days in the table.)

Day of class	Shakir (wpm)	Julia (wpm)
1st	25	35
49th	45	40
99th	55	55

53. To the nearest 1%, what is the percent increase in Julia's typing speed from the 1st day of class to the 99th day of class?
- A. 14%
 - B. 20%
 - C. 36%
 - D. 57%
 - E. 64%
54. Which of the statements below, if any, *must* be true about Julia's typing speed from the 49th day through the 99th day?
- I. Julia's typing speed on at least one day was 50 wpm.
 - II. Julia's typing speed on at least one day was increased by 1 wpm from the previous day.
 - III. Julia's maximum typing speed during the 99-day class was 55 wpm.
- F. I only
 - G. II only
 - H. III only
 - J. I, II, and III
 - K. None
55. Which of the following values is closest to the average rate of change of Shakir's typing speed, in words per minute per day, from the 1st day of class to the 99th day of class?
- A. 0.202
 - B. 0.306
 - C. 3.3
 - D. 15
 - E. 20



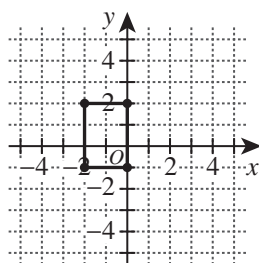
56. Rectangle $ABCD$ is graphed in the standard (x,y) coordinate plane below.



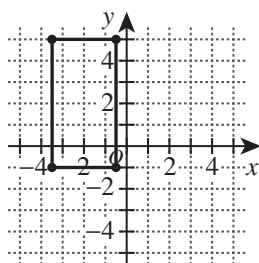
DO YOUR FIGURING HERE.

Rectangle $ABCD$ undergoes a dilation with a center of A and a scale factor of $\frac{3}{2}$. One of the following is the graph of the image of rectangle $ABCD$ under this dilation. Which one?

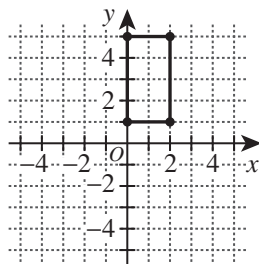
F.



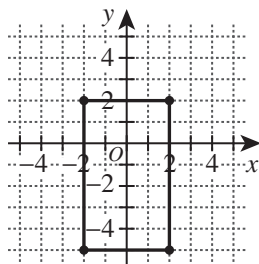
J.



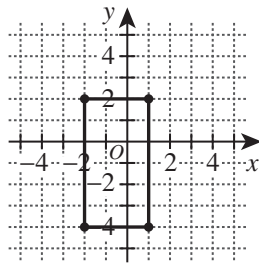
G.



K.



H.

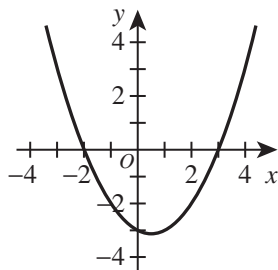


57. A straight line through the point with (x,y) coordinates $(-1,1)$ has slope 2 and contains the point $(3,d)$. What is the value of d ?

- A. 3
- B. 4
- C. 7
- D. 8
- E. 9



58. A parabola that contains the points $(0,-3)$, $(-2,0)$, and $(3,0)$ is shown in the standard (x,y) coordinate plane below.



The function f has the graph of this parabola translated 3 coordinate units right. Which of the following is the output of $f(x)$?

- F. $-\frac{1}{12}(x-9)(x-3)$
- G. $\frac{1}{2}(x-6)(x-1)$
- H. $\frac{1}{2}(x-5)(x)$
- J. $\frac{1}{2}x(x+5)$
- K. $\frac{1}{2}(x+1)(x+6)$
59. In the standard (x,y) coordinate plane, an angle with measure α is in standard position, and the point $(-3,7)$ is on the terminal side of the angle. What is $\cos \alpha$?
- A. $-\frac{3}{7}$
- B. $-\frac{3}{\sqrt{58}}$
- C. $-\frac{7}{\sqrt{58}}$
- D. $\frac{3}{7}$
- E. $\frac{3}{\sqrt{58}}$
60. The real numbers a and b are unknown. It is known that $a < 0$, $b < 0$, and $\frac{a}{b} < 1$. Therefore, $a - b$ must be:
- F. less than 0.
- G. less than or equal to 0.
- H. less than or equal to 1.
- J. greater than 0.
- K. greater than or equal to 1.

DO YOUR FIGURING HERE.

END OF TEST 2

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO THE PREVIOUS TEST.

READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are several passages in this test. Each passage is accompanied by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

LITERARY NARRATIVE: This passage is adapted from the novel extract “Sparks” by Carol Bensimon (©2012 by Granta Publications).

The passage is set in Porto Alegre, the largest city in the Brazilian state of Rio Grande do Sul.

My car had been out of action for some time, under a silver waterproof cover, like a big secret you just can’t hide or a child trying to disappear by putting her hands over her eyes, surrounded by junk in the garage at my mother’s house. Initially, my mother was desperate to resolve the situation. It’s a bad business leaving a car off the road for so long, she would say, although she understood very little about business and even less about getting rid of things. She lived in a house that already seemed too big when there were still three of us. When you opened certain wardrobes in that house, you could see the entire evolution of ladieswear from the mid-sixties onwards. Lovely jackets, pretty dresses that didn’t fit my mother any more. I was direct about the car. I said: ‘Maybe I’ll come back.’ I could sense her breath crossing the ocean and almost capsizing before returning to dry land. Perhaps it was a mistake to offer hope to a single mother, given that I wasn’t even considering the possibility of moving home at that point. We never spoke about the car again.

Three years later, I was back and found the garage fuller than ever, so much so that I could barely see the terracotta floor tiles for the bags full of papers, the boxes of all sizes. There were balls of dust, an electric heater, a small bicycle, a minibar missing a leg. I got the impression I could have written WASH ME in the air with my index finger. I pushed open the wooden concertina doors and let in the light. I stood looking at the street for a while. It was no longer the same street, I mean, it was the same street, but in place of the houses belonging to my childhood friends—where were they now?—an apartment block had been built. It scared me to think that one person’s aesthetic preferences could be summed up in that white, seventeen-floor mastodon.

Apart from that, there were other subtle changes to that section of the street. They did not date from the last three years, however, the three years I had spent away from Porto Alegre and that house, during which time I had rarely imagined my return and the exhausting list

40 of comparisons that would almost certainly stem from it. For some reason, what I was trying to do was rebuild the street of my adolescence and my difficulty in achieving that made me think about those little books you get in Rome, Italy, where, by superimposing two images, you can see something that was once grandiose where now there are only remains of columns, marble blocks or a sizeable area of grass.

Then I went back into the garage. I pulled the cover off the car. It was very clean. A strange, metallic blue body in the midst of all that dusty chaos. The battery, though, or whatever it was, had gone to pot.

Even though the car was unfit to drive right then, I adjusted the back of the seat and stayed sitting there. I very nearly put my hands on the wheel. But cars weren’t my obsession. You ask me what model just went past and I’ll never be able to say. It was their mobility that appealed to me, mobility as an end. And I was thinking how obscure that is when you are first presented with a car, how, at eighteen, with your driving license in a flawless plastic sheath and that ridiculous photo with the haircut you’ll regret later, all you want to do is cruise along open roads at dawn without ever getting anywhere. Or rather, your anywhere is an album to be heard in full, with as many friends as you can fit in the back seat. The curious thing is that keeping these habits beyond their expiry date makes them seem, in the eyes of others, to be nothing more than a sign of eccentricity in someone who never knew how to grow up.

My mother entered the garage as I was reminiscing. In the rearview mirror, I saw her running her fingers over the dust-covered boxes, head bowed, giving the impression that she was reading what might be written there, as if until that moment she had ignored their contents or didn’t even know why they were piled up in her garage. I got out of the car and waited for her to approach. She gave me one of her out-of-context smiles. ‘Won’t it start, Cora?’ It was quite common for bad news to come out of my mother’s mouth accompanied by a smile. Not out of spite, quite the opposite; there was some notion of compensation in it.

‘I think it would have been a miracle if it had,’ I said.

We agreed that it probably wasn't anything serious, nothing a mechanic couldn't sort with the turn of a wrench. We stayed standing there. I looked around me. Funny I couldn't remember that tiny bicycle. No one other than me had been a child in this house.

1. The point of view from which the passage is told is best described as that of a:
 - A. third person limited narrator who primarily relates Cora's thoughts about a series of events that happened in the past.
 - B. third person omniscient narrator who primarily reports on Cora's and Cora's mother's thoughts and actions as they occur.
 - C. first person narrator who primarily reports on her mother's actions as they occur.
 - D. first person narrator who primarily relates her thoughts surrounding an event that happened in the past.
2. What is the main theme of the passage?
 - F. Driving feels more satisfying at the age of eighteen than it does at any other age.
 - G. No matter how many objects a person saves, he or she will not be able to accurately remember the past.
 - H. Distance from a once well-known place invites processing and conceptualizing that place upon returning to it.
 - J. A parent-and-child bond never feels as strong as when the child arrives home after a long absence.
3. Based on the passage, Cora can most nearly be characterized as:
 - A. underhanded and vengeful.
 - B. ruminative and analytical.
 - C. carefree and whimsical.
 - D. empathetic and hardworking.
4. In the passage, Cora suggests that an adult holding onto the youthful habit of driving without a destination is seen by others as:
 - F. peculiar and immature.
 - G. adventuresome and bold.
 - H. intellectual and worldly.
 - J. wasteful and dull.
5. In the passage, Cora's mother's actions when first entering the garage can best be described as an attempt to:
 - A. appear busy as she waits to approach Cora.
 - B. determine whether Cora is in the garage.
 - C. investigate a corner of the garage that she has typically ignored.
 - D. read and memorize what was written on the dust-covered boxes.
6. In the context of the passage, Cora's statement "I could sense her breath crossing the ocean and almost capsizing before returning to dry land" (lines 15–17) most nearly describes Cora's mother literally:
 - F. traveling across an ocean.
 - G. whispering softly.
 - H. exhaling deeply.
 - J. nearly falling down.
7. Which of the following statements provides the best paraphrase of the sentence in lines 32–34?
 - A. Cora was dismayed to realize that, having the choice of something to build, someone would choose to create something so big and plain.
 - B. Cora was disheartened to learn that a second apartment block would be built at the former site of one of her childhood friend's homes.
 - C. Cora was panicked to discover that most people's aesthetic preferences were unlike her own.
 - D. Cora was frightened to see that one huge, white building could define an entire street.
8. It can most reasonably be inferred from the passage that compared to the condition of the inner workings of Cora's car at the time Cora uncovers it, the condition of the car's exterior was:
 - F. much better.
 - G. just as mediocre.
 - H. slightly worse.
 - J. much worse.
9. In the passage, what main reason does Cora provide to support her claim that cars are not her obsession?
 - A. She has not owned a car during the three years that she has been gone from Porto Alegre.
 - B. She did not care whether her car was covered while it was stored in her mother's garage.
 - C. She did not choose to put her hands on the steering wheel when she sat in her car.
 - D. She would not be able to name any given model of car that drives by.
10. As it is used in line 81, the phrase "there was some notion of compensation in it" most nearly suggests that Cora believes her mother smiled when giving bad news to:
 - F. appear to be telling the truth.
 - G. soften the message.
 - H. playfully tease.
 - J. show spite.

Passage II

SOCIAL SCIENCE: Passage A is from the book *Dry Storeroom No. 1: The Secret Life of the Natural History Museum* by Richard Fortey. Passage B is from the article “Hollywood Knights” by Tom McNichol.

Passage A by Richard Fortey

Ron Croucher is a most self-effacing kind of man, but I believe he has little doubt that *Baryonyx walkeri* is his masterpiece. His apprenticeship as a preparator went back to the early days of the Palaeontology Department. In the old Museum the “lab” was housed in dreadful conditions at the north side of the building. It was so cold in winter that a series of electric fires were propped all over the place to keep the staff warm, and the lighting was so poor that it was sometimes necessary to wait for the sun to creep into the room before a particularly delicate task could be undertaken. Nonetheless, innovative work was carried out there. The staff pioneered the use of resins in casting fossils, which is now routine. Ron Croucher’s predecessor, Arthur Rixon, pioneered the extraction of fishes from limestone by acid solution. A fish—like us—has bones of calcium phosphate, whereas limestone is calcium carbonate; the latter dissolves in certain acids, acetic acid perhaps being most commonly employed, leaving the bones untouched. The result can be the most exquisite preparations of fossils, as precise as a dissection, with no tool brought to bear on the fragile skeleton.

No such short cuts were possible in the case of *Baryonyx*. Ron had to cut off the sandstone surrounding the bones grain by grain using a vibrating air pen with a hardened tip. It is painstaking work: a moment’s lapse in concentration can polish off the work of days. In the case of *Baryonyx*, extraction was even more difficult, because the bones were softer than the surrounding rock. Certainly much more than a year of Ron’s life went into the work; he says that he lost all sense of time after a while. The final result is a reconstruction of the dinosaur as a specialized fish eater, a predator with an almost crocodile-like skull and huge claws; it is related to *Spinosaurus*, but tells a new story about the adaptability of the ruling reptiles to different life habits. Ron’s contribution was acknowledged in the scientific description of *Baryonyx* by Alan Charig and Angela Milner—a formal nod towards his hours and hours of patient toil. When dinosaurs appear in books and films as realistic as if they had been plucked from the Cretaceous by a time machine, it is easy to forget that everything we know is the result of the labours of unsung masters like Ron Croucher. Reality is extracted out of sight of the public in back rooms full of half-exposed bones.

Passage B by Tom McNichol

During production of the animated film *Coraline*, nearly 150 sets were spread out in a cavernous warehouse, each a meticulous miniature world in which the animators staged the action. One set was so intricate it featured a floor that collapsed into a spider web into which Coraline tumbles. The two-story set used computer-controlled winches to make the thin wire web collapse in just the right pattern.

At the height of production, about 450 animators, designers, modelers, sketch artists, carpenters, and camera people swarmed through the warehouse, building *Coraline* frame by frame. Each upraised eyebrow, painted fingernail, and blade of grass in the movie exists in real life, albeit in miniature. The real “actors” are the animators who manipulated the puppets. “It’s exhausting work,” says lead animator Travis Knight. “It’s a performance in slow motion that you act through the puppet. You’re burning your fingers on hot glue, you’re cutting your hands on bits of armature wire, you’re crawling through sets and banging your head on pipes. You go home and you feel like you’ve been digging trenches for ten hours. The only reason stop-motion characters have life is because they take it out of someone else. They’re little vampires that suck the life out of the people who work on them.”

Coraline employed thirty-five animators in all, some recruited from Disney and DreamWorks, some lured from England; still others were local recruits. To shoot each scene, animators had to pose the puppets, shoot a single frame of film, stop the camera, carefully change the position of the puppets and scenery to imply motion, and then shoot another frame. To complete one second of finished film, this procedure had to be repeated twenty-four times. “I wouldn’t do what Travis is doing for anything,” says CEO of the animation company Phil Knight, shaking his head. “I would say to him, ‘How was your day?’ and he would say, ‘Oh good, I got ten seconds in the can.’ And I’d think, ‘Ten seconds—for a whole day!’ It would drive me crazy.”

Passage A: Excerpt(s) from DRY STOREROOM NO. 1: THE SECRET LIFE OF THE NATURAL HISTORY MUSEUM by Richard Fortey, copyright ©2008 by Richard Fortey. Used by permission of Alfred A. Knopf, an imprint of the Knopf Doubleday Publishing Group, a division of Penguin Random House LLC. All rights reserved.

Passage B: From the article “Hollywood Knights” by Tom McNichol (©2012 by Tom McNichol). Tom McNichol is a San Francisco writer. Used with permission.

11. The main purpose of Passage A is to:
- A. describe the different ways in which fossils were prepared at the museum.
 - B. pay homage to a particular preparator by focusing on one of the fossils he extracted.
 - C. point out the benefits and drawbacks of working as a preparator at the museum.
 - D. provide an example of an extracted fossil that a preparator immortalized in film.
12. In Passage A, details about the lab in the Palaeontology Department in part serve to:
- F. demonstrate what Croucher and his colleagues enjoyed most about working at the museum.
 - G. indicate why it was difficult for the museum to attract the top paleontologists in the field.
 - H. highlight the disparity between the poor conditions in which the staff worked and the valuable innovations they developed.
 - J. demonstrate the similarities between where the staff worked in the “early days” and where they work currently.
13. According to Passage A, the extraction of the *Baryonyx* fossil was particularly challenging because the:
- A. museum had limited access to the tools needed to prepare the fossil.
 - B. fossil was missing key parts.
 - C. limestone around the fossil dissolved after being exposed to acetic acid.
 - D. bones of the fossil were softer than the rock surrounding it.
14. It can reasonably be inferred from Passage A that part of what makes the extracted *Baryonyx* fossil a “masterpiece” is the:
- F. quickness with which Croucher completed the work.
 - G. revenue potential of the piece.
 - H. size of the work.
 - J. precision of Croucher’s work.
15. The main point of the quotation from Travis Knight in lines 61–71 is that:
- A. working as a stop-motion animator garners instant celebrity in the film industry.
 - B. working on the set of a stop-motion film is grueling.
 - C. people outside the film industry are fascinated by stop-motion animation.
 - D. people outside the film industry think that stop-motion films are easy to make.
16. According to Passage B, how many animators were employed for *Coraline*?
- F. 24
 - G. 35
 - H. 150
 - J. 450
17. As he is presented in Passage B, Phil Knight reacts to Travis Knight’s supposed comment “Oh good, I got ten seconds in the can” (lines 83–84) with what can best be described as:
- A. incredulity.
 - B. reverence.
 - C. disapproval.
 - D. scorn.
18. Based on the passages, the *Baryonyx* and *Coraline* projects were similar in that they both required:
- F. extensive collaboration throughout the creative process.
 - G. broad experience with a variety of art forms.
 - H. meticulous attention to detail.
 - J. specialized knowledge of biology.
19. Which of the following quotations from Passage A or Passage B best encapsulates a central theme of both passages?
- A. “The result can be the most exquisite preparations of fossils” (lines 20–21).
 - B. “Reality is extracted out of sight of the public in back rooms” (lines 44–45).
 - C. “The two-story set used computer-controlled winches” (lines 52–53).
 - D. “I wouldn’t do what Travis is doing for anything” (lines 80–81).
20. The details about cutting sandstone in Passage A and repositioning puppets and scenery in Passage B are similar in that they help to:
- F. identify the number of hours it took to complete the project.
 - G. provide information about the costs incurred during the project.
 - H. describe a technical process employed throughout the project.
 - J. specify the traditional tools used for the project.

Passage III

HUMANITIES: This passage is from the book *The Contemporaries: Travels in the 21st-Century Art World* by Roger White.

You could begin with the *bottegas*, where a successful artist of the Renaissance might have employed dozens of apprentices to help with all aspects of his craft. These workshops occupied a place at the center of a community. Pre-adolescents—not necessarily artistically inclined ones, just neighborhood kids who needed work—progressed from simple tasks to more challenging ones: mixing up batches of gypsum and glue for use as a painting ground, then transferring the master's designs onto wooden panels or altar walls, then grinding pigments and making paints, and then, years later, executing parts of actual paintings.

Each workshop was a small business and a school in miniature. In Florence, they predated both the formation of a dedicated guild for artists and an academy of the arts. Each had its own way of doing things, and this proprietary technique, inseparable from the meaning of the art it produced, was sometimes even written down, ensuring the faithful transmission of the artist's method. Ambitious assistants aimed to graduate from service and eventually hang their own shingles in town, competing with their former employers for commissions from churches and nobles, and in turn training the next generation of artists, world without end.

The workshop system also explains the way artistic influence propagated in Renaissance Italy: through imitation. At the heart of the apprenticeship process was the act of copying. Assistants learned by reproducing the master's drawings, paintings, or sculptures (which, more often than not, were reproductions of other artists' drawings, paintings, or sculptures) with the aim of perfect fidelity to the boss's style. This practice had the added benefit of generating a stockpile of reference material, closely guarded by the workshop, for use in future projects: *You need a horse and rider, two cherubs, and Our Lady seen in left profile? No problem, Maestro. I'll pull them from the file.* The Florentine painter Domenico Ghirlandaio kept bound volumes of drapery studies (drapes at rest, drapes in a gentle breeze, drapes wrapped austerely around an apostle or opulently around a wealthy patron) to deploy in his commissions, like Renaissance clip art.

But beyond the creation of these visual reference libraries, the practice of copying prepared the assistant for executing parts of actual artworks, according to the master's specifications. Once the assistant's work was up to snuff, he could be tasked with completing the tedious bits of a commission: some shrubbery along the road to Calvary, or the wings of the lesser angels in a Last Judgment. Certain very talented apprentices, it was said, couldn't help but reveal their employers' limitations by copying too well—in the way the precocious Raphael once outshone his master Perugino on an altarpiece job the latter artist was foolish enough to assign to the former.

By 1610, the Flemish painter Peter Paul Rubens had internationalized the workshop model, with teams of protégés in his Antwerp studio executing works for patrons from London to Madrid, while the artist himself traveled around hobnobbing with potential patrons and, in fact, performing diplomatic missions for the Spanish Hapsburg court; he was knighted, twice.

The French Academy in the eighteenth century then centralized it, linking art training, productions, exhibition, and patronage in a monolithic, state-sponsored machine. The most successful artists of the period maintained extensive studios, with assistants drawn from the ranks of academy students, to meet the considerable demands of royal and republican commissions.

As the influence of the French Academy waned and state patronage dried up, so too did the collaborative mode of the workshop. The new, private patron of the arts, drawn from the upper tiers of the bourgeoisie, bought modestly sized pictures for the home. The Salon des Refusés of 1863, an exhibition organized for French artists turned away from the Académie des Beaux-Arts' annual show, marked a watershed in the aesthetics of artistic success. Manet, Whistler, and Cézanne debuted works there, to great public scandal and belated acclaim; in hindsight, it was better to have been rejected from the official Salon of 1863 than accepted by it. With this shift in the structure of patronage, the social coordinates of the artist were dramatically relocated: from the comfortable center of high-cultural life to its adventurous margins.

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21. The main purpose of the passage is to:
 - A. compare Italian Renaissance artists' styles to those of artists from the French Academy.
 - B. suggest that the importance of art from the Italian Renaissance has been overestimated by art historians.
 - C. contend that art apprentices in Renaissance Italy and eighteenth-century France were more creative than has been supposed.
 - D. describe Italian Renaissance artists' workshops and their effects on the production of art in Europe.
22. It can most reasonably be inferred from the passage that an apprentice's best chance for advancement in an artist's workshop in Renaissance Italy was to:
 - F. join a dedicated artists' guild.
 - G. gain acceptance to an academy of the arts.
 - H. master the artist's style.
 - J. paint with originality.

23. According to the passage, during the Italian Renaissance, the first task an apprentice was asked to complete was:
- A. transferring designs onto wooden panels.
 - B. mixing batches of gypsum and glue.
 - C. grinding pigments.
 - D. making paints.
24. The author most likely compares Ghirlandaio's drapery studies to clip art in order to:
- F. indicate the ease with which Ghirlandaio could incorporate the drapery studies into his commissioned work.
 - G. suggest the artistry in the drapery studies was crude compared to Ghirlandaio's other works of art.
 - H. reveal why Ghirlandaio's paintings appear uncannily modern.
 - J. explain how Ghirlandaio's apprentices enhanced Ghirlandaio's color palette.
25. In the context of the passage, the author refers to Rubens mainly in order to:
- A. indicate that he was knighted twice.
 - B. reveal that he was a key figure in the spread of the workshop model.
 - C. describe how he created the first academy for the arts.
 - D. document how he set up multiple studios throughout Europe.
26. Based on the passage, the general public's initial reaction to the Salon des Refusés exhibition of 1863 was most nearly one of:
- F. admiration.
 - G. sadness.
 - H. shock.
 - J. tolerance.
27. In the passage, the statement that "it was better to have been rejected from the official Salon of 1863 than accepted by it" (lines 81–82) can best be characterized as:
- A. a reasoned judgment put forth by the author based on a recognition of a structural shift in art patronage.
 - B. a reasoned judgment put forth by the author based on an understanding of the far-reaching influence of the workshop model.
 - C. an opinion held by private patrons who preferred Italian Renaissance art to French art.
 - D. an opinion held by private patrons who were wary of art created by adventurous artists.
28. In the passage, the author most strongly suggests that the art produced by a workshop in Renaissance Italy was most directly influenced by the:
- F. age of the workshop's apprentices.
 - G. creativity of a workshop's apprentices.
 - H. workshop's size.
 - J. workshop's proprietary technique.
29. Based on the passage, the italicized text in lines 35–37 is most likely meant to reflect the perspective of an unidentified:
- A. patron.
 - B. art appraiser.
 - C. apprentice.
 - D. art critic.
30. In the passage, the author most likely uses the phrase "copying too well" (line 52) to emphasize that:
- F. Raphael's execution of the altarpiece job was more artistically accomplished than Perugino's.
 - G. Raphael's execution of the altarpiece job was indistinguishable from Perugino's.
 - H. Perugino demanded that his apprentices spend long hours honing their craft.
 - J. Perugino became a champion of Raphael's artistic endeavors.

Passage IV

NATURAL SCIENCE: This passage is from the book *The Homing Instinct: Meaning and Mystery in Animal Migration* by Bernd Heinrich.

Altricial refers to species in which the young are born or hatched in a very immature and helpless condition. *Precocial* refers to species that, from birth, are capable of a high degree of independent activity.

A fortress, if it is impregnable, does not need to be hidden. Cavities, especially those in trees, are one of the most common ready-made fortresses. There are few major bird groups in which some members do *not* avail themselves of this resource. The problem is that it is often a limited resource, and depending on this scarce resource may severely limit reproductive opportunity. Therefore, the ultimate in home-making for hole nesters is to excavate their own home, to their own specifications. Woodpeckers, perhaps because they are pre-adapted for it because of their foraging for wood-boring grubs, are the experts at it.

Woodpeckers excavate their own home cavities out of solid wood, with the entrance reduced to a minimum size that excludes almost all predators larger than they are. Small predators that might be able to squeeze in can likely be repelled with the same tool used to make the home, the sharp and powerful bill. Larger predators that can't enter might still be able to reach in through the hole, but if the cavity is made deep enough, they can't reach the eggs and young on the bottom. Woodpeckers may expend what appears to be extraordinary effort to make a home, but the effort is worth it, because they suffer virtually no nest predation.

Woodpeckers make no secret of where their homes are. In June when they have young, I can locate a woodpecker nest from within a hundred meters, whereas I may step over any of a variety of other forest bird nests without noticing them. A woodpecker comes and goes to its nest at frequent intervals, and the young in the nest make a nonstop din that increases in volume whenever they sense something landing on their home tree. All other forest bird babies stay silent until a parent is almost directly at the nest, and then the babies peep only faintly and only the moment the parent arrives with food. They shut up the moment it leaves. There is never (in most species) a single fecal droplet in or near the nest due to a complex interactive behavior between the parents and their offspring. In contrast, with the woodpecker nest, after the older young start to monopolize the nest entrance, feces accumulate and are simply trodden down into debris that collects on the floor. There is, it appears, a downside to almost everything, and each solution is a balance of trade-offs.

Similar scenarios of homing apply in mammals as well, and the two divergent strategies of altricial versus precocial young are exhibited in closely related species. Those of the hares and rabbits, Leporidae, and apes and humans, Hominidae, are the most interesting and instructive examples.

Hares and rabbits look very similar to our eyes but have vastly different life and home strategies. Hares, *Lepus*, such as jackrabbits and varying or snowshoe hares, bear their young into a simple bare depression on the ground. They have no protective home but, like precocial birds, are born with insulating and camouflaging covering. Like baby sandpipers, ducks, chickens, deer, and antelope, they are able to run from birth. Rabbits, on the other hand, are born blind and naked into a furlined nest (the hair the mother has plucked from her body). One species, the European rabbit, *Oryctolagus cuniculus*, is the ancestor from which all domesticated rabbits are derived, and in the wild, this species digs networks of tunnels to create what is called a "warren." One might suppose that living in an excavated cavity is safer than having the young deposited in the open on the ground. It could be, but an earthen tunnel is not impregnable. Some predators have evolved to enter tunnels. A rabbit warren must have many potential escape exits. But it has an additional design feature—a nursery chamber with the nest and babies in a side tunnel that is without an exit. This is not a design flaw but a "clever" safety feature, since the helpless young could not escape in any case. Rather than allowing a predator that searches within a warren to reach the nest from two directions, there is only one. Furthermore, this entrance is plugged most of the time, and the doe (female rabbit) reduces the time window when the tunnel to the nest is open to a bare minimum—the three or four minutes per day, at almost exactly the same time every twenty-four hours, that she visits to nurse them. The young learn when to expect her and get ready by uncovering themselves to be ready to nurse the moment she comes in. After having nursed her babies, she leaves immediately and plugs the nest entrance behind her.

Excerpt from *THE HOMING INSTINCT: Meaning and Mystery in Animal Migration* by Bernd Heinrich. Copyright © 2014 by Bernd Heinrich. Reprinted by permission of Houghton Mifflin Harcourt Publishing Company. All rights reserved.

31. The passage makes clear that one similarity between the woodpecker and the European rabbit is that both animals:
 - A. repel nest predators with their own beaks or teeth.
 - B. cover the opening of their nests most of the time.
 - C. line their nests with their own feathers or fur.
 - D. excavate their own homes.
32. According to the passage, one difficulty for birds in using preexisting tree cavities for their homes is that the cavities are:
 - F. easily located by snakes and weasels.
 - G. unpredictable in size.
 - H. often rotted.
 - J. scarce.

33. The main idea of the second paragraph (lines 13–24) is that:
- A. larger predators can't enter a woodpecker nest but still might be able to reach in through the hole.
 - B. the woodpecker creates a home out of solid wood and makes a small entrance.
 - C. predators tend to ignore a woodpecker nest knowing that the eggs would be hard to acquire.
 - D. the woodpecker creates a home in which eggs and young are almost completely safe from predators.
34. What evidence does the author provide to support the claim that "woodpeckers make no secret of where their homes are" (lines 25–26)?
- F. There appears to be a downside to the woodpecker's homemaking strategies.
 - G. A young woodpecker can sense when something has landed on its home tree.
 - H. A woodpecker comes and goes from its nest frequently.
 - J. It's easy not to notice the nests of many forest birds.
35. It can reasonably be inferred from the passage that which of the following animals is least independent from birth?
- A. European rabbit
 - B. Sandpiper
 - C. Duck
 - D. Hare
36. What reason does the author provide to support his claim that the lack of exit on a nursery chamber tunnel in a rabbit warren isn't a design flaw but a "clever" safety feature" (lines 72–73)?
- F. An exit would take considerable energy for a doe to create, but an exit wouldn't provide added protection.
 - G. The young couldn't escape even if there were an exit, but an exit could increase the possibility that a predator will find the nest.
 - H. Although an exit would let fresh air into the nursery chamber, it would also present the risk of flooding.
 - J. Although an exit might provide the young an escape route, it would also allow them to leave the warren prematurely.
37. In the passage, which of the following structures does the author most directly compare to a fortress?
- A. A depression in the ground
 - B. A cavity in a tree
 - C. An earthen tunnel
 - D. A nursery chamber
38. As it is used in line 15, the word *excludes* most nearly means:
- F. prohibits.
 - G. removes.
 - H. suspends.
 - J. ejects.
39. Based on the passage, to ensure the safety of the eggs and young in a woodpecker nest, the nest must be sufficiently:
- A. deep.
 - B. narrow.
 - C. high up in a tree.
 - D. camouflaged in a tree.
40. The author asserts that hares and rabbits provide a particularly good example that supports which of the following claims?
- F. Closely related species may sometimes illustrate the very different home strategies of altricial and precocial young.
 - G. The home strategies of some small mammals may mirror the home strategies of apes and humans.
 - H. A species may quickly evolve to adapt the fruitful homemaking habits of a closely related species.
 - J. Precocial species may evolve to survive a range of threats to their homes, whereas altricial species typically remain vulnerable.

END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO A PREVIOUS TEST.



SCIENCE TEST

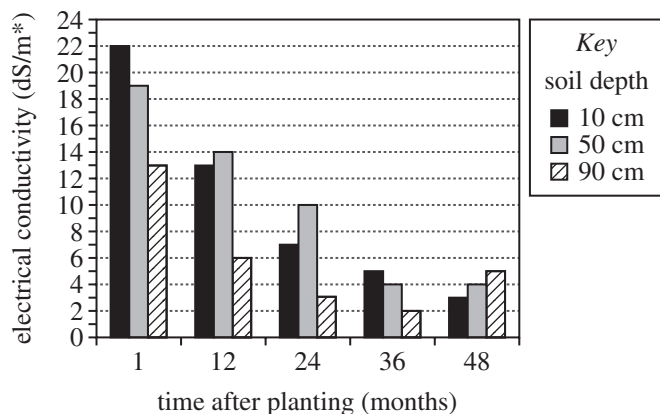
35 Minutes—40 Questions

DIRECTIONS: There are several passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage I

Kallar grass (a salt-tolerant grass) was planted in a soil having a high salt content. As the roots of the grass grow and release CO_2 , the soil pH decreases. Ions, such as calcium (Ca^{2+}), from dissolved salts may then be more readily *leached* (carried downward by water) out of the upper soil layers. Figure 1 and Table 1 show the *electrical conductivity* (a measure of the number of dissolved ions in the soil that is directly proportional to salt content) and the soil Ca^{2+} concentration, respectively, for each of 3 soil depths at 5 different times after planting.



*decisiemens per meter

Figure 1

Table 1			
Time after planting (months)	Ca^{2+} concentration (in meq/L*) at a soil depth of:		
	10 cm	50 cm	90 cm
1	3.7	2.6	2.0
12	2.0	2.1	1.8
24	1.2	1.4	0.4
36	0.9	0.7	1.2
48	0.6	1.0	0.5

*milliequivalents per liter

Figure 1 and Table 1 adapted from Akhter, J. et al., "Amelioration of a Saline Sodic Soil through Cultivation of a Salt-Tolerant Grass *Leptochloa fusca*." ©2003 by Foundation for Environmental Conservation.

- Based on Table 1, at 36 months after planting, which soil depth—10 cm or 50 cm—had the *lower* Ca^{2+} concentration?
 - 10 cm; the Ca^{2+} concentration at a soil depth of 10 cm was 0.2 meq/L less than that at a soil depth of 50 cm.
 - 10 cm; the Ca^{2+} concentration at a soil depth of 10 cm was 0.4 meq/L less than that at a soil depth of 50 cm.
 - 50 cm; the Ca^{2+} concentration at a soil depth of 50 cm was 0.2 meq/L less than that at a soil depth of 10 cm.
 - 50 cm; the Ca^{2+} concentration at a soil depth of 50 cm was 0.4 meq/L less than that at a soil depth of 10 cm.



2. Based on Figure 1, the electrical conductivity for a soil depth of 50 cm at 1 month after planting was approximately how much greater than the electrical conductivity for a soil depth of 10 cm at 36 months after planting?
- F. 7 dS/m
G. 9 dS/m
H. 12 dS/m
J. 14 dS/m
3. According to Table 1, at a soil depth of 50 cm, between which of the following times after planting did the Ca^{2+} concentration increase?
- A. Between 1 month and 12 months
B. Between 12 months and 24 months
C. Between 24 months and 36 months
D. Between 36 months and 48 months
4. In the study, was electrical conductivity a dependent variable or an independent variable, and was time after planting a dependent variable or an independent variable?
- | | electrical conductivity | time after planting |
|----|-------------------------|---------------------|
| F. | dependent | dependent |
| G. | dependent | independent |
| H. | independent | dependent |
| J. | independent | independent |
5. Consider the statement “At 12 months after planting, the soil depth that had the *lowest* electrical conductivity was also the depth that had the *lowest* Ca^{2+} concentration.” Is this statement consistent with the data shown in Figure 1 and Table 1 ?
- A. Yes; at 12 months after planting, the electrical conductivity and the Ca^{2+} concentration were both the lowest at a depth of 50 cm.
B. Yes; at 12 months after planting, the electrical conductivity and the Ca^{2+} concentration were both the lowest at a depth of 90 cm.
C. No; at 12 months after planting, the electrical conductivity was lowest at a depth of 50 cm, but the Ca^{2+} concentration was lowest at a depth of 90 cm.
D. No; at 12 months after planting, the electrical conductivity was lowest at a depth of 90 cm, but the Ca^{2+} concentration was lowest at a depth of 50 cm.
6. Consider the claim “As the soil depth increased from 10 cm to 50 cm and then to 90 cm, the Ca^{2+} concentration increased and then decreased.” This claim is consistent with the data shown in Table 1 for which of 1 month and 24 months after planting, if either?
- F. 1 month only
G. 24 months only
H. Both 1 month and 24 months
J. Neither 1 month nor 24 months



Passage II

Heavy metals in the environment often have concentrations too low to be detected by common testing methods. *Solid phase extraction* (SPE) can be used to improve heavy metal detection.

Scientists performed 3 SPE experiments to examine the use of graphene (a form of pure carbon) in the analysis of cadmium (a heavy metal) in aqueous samples.

Experiment 1

In each trial, a different mass of graphene was added to 0.05 L of a 0.1-micrograms per liter ($\mu\text{g/L}$) cadmium solution of a specific pH. This sample was incubated in a water bath for 40 min at a specific temperature. The sample was removed from the bath, and the graphene was separated from the liquid by centrifugation (spinning). The isolated graphene was treated with a solution to release the cadmium from the graphene, and then the percent recovery of cadmium was determined. See Figure 1.

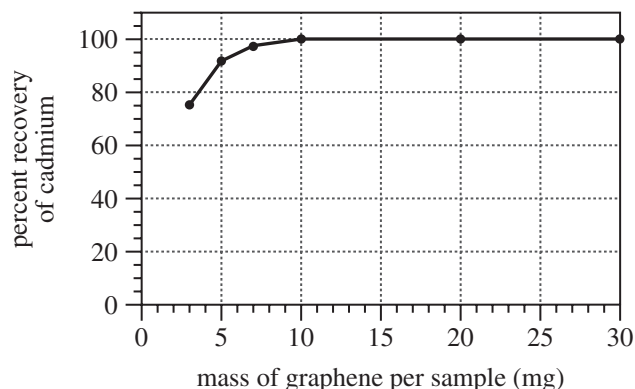


Figure 1

Experiment 2

Experiment 1 was repeated except that an identical mass of graphene was added to the cadmium solution in each trial and each sample had a different pH. See Figure 2.

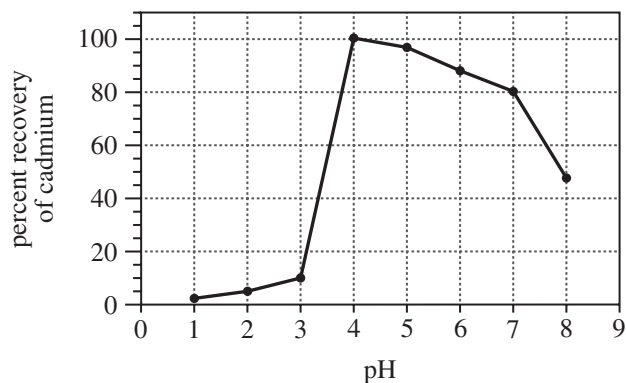


Figure 2

Experiment 3

Experiment 1 was repeated except that an identical mass of graphene was added to the cadmium solution in each trial and each sample was incubated at a different temperature. See Figure 3.

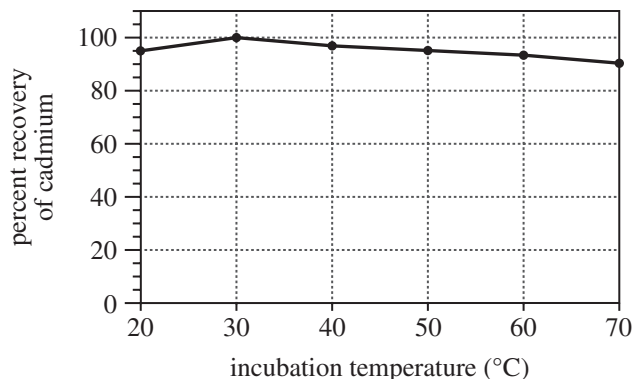


Figure 3

Figures adapted from R. Hallaj et al., "Adsorption of Cadmium(II) Ions from Aqueous Solution on Exfoliated Graphene Nanosheets and Its Determination by Flame Atomic Absorption Spectrometry." ©2014 by NRC Research Press.

7. Based on the description of Experiment 1, when was the graphene separated from the sample liquid by centrifugation?
 - A. Immediately before placing the sample in the water bath
 - B. Immediately after removing the sample from the water bath
 - C. Immediately before the percent recovery of the cadmium was determined
 - D. Immediately after the percent recovery of the cadmium was determined
8. Suppose that in Experiment 2 a solution at a pH of 7.5 had been tested. Based on the results of Experiment 2, the percent recovery of cadmium for that solution would most likely have been:
 - F. between 48% and 80%.
 - G. between 80% and 88%.
 - H. between 88% and 95%.
 - J. between 95% and 100%.
9. Based on the results of Experiment 1, when 5.0 mg of graphene was added to the solution, the percent recovery of cadmium was closest to which of the following values?
 - A. 70%
 - B. 80%
 - C. 90%
 - D. 100%



10. Based on the results of the 3 experiments, which of the following sets of conditions (for mass of graphene, solution pH, and bath temperature) would most likely result in 100% recovery of the cadmium?
- F. 5.0 mg of graphene, pH of 4, water bath temperature of 30°C
 - G. 10.0 mg of graphene, pH of 4, water bath temperature of 30°C
 - H. 5.0 mg of graphene, pH of 7, water bath temperature of 40°C
 - J. 10.0 mg of graphene, pH of 7, water bath temperature of 40°C
11. Assume that each data point in Figure 1 represents 1 trial. Based on the description of Experiment 1, if the sample for each trial was incubated separately, how much *total* time were there samples in the water bath?
- A. 40 min
 - B. 50 min
 - C. 240 min
 - D. 320 min
12. A scientist claimed that as the mass of graphene added to a cadmium solution was increased from 3.0 mg to 30.0 mg, the percent recovery of cadmium would increase linearly until it reached 100%. Do the results of Experiment 1 support this claim?
- F. Yes; as the mass of added graphene increased, the percent recovery increased linearly, reaching 100% when 1.0 mg of graphene was added.
 - G. Yes; as the mass of added graphene increased, the percent recovery increased linearly, reaching 100% when 2.0 mg of graphene was added.
 - H. No; as the mass of added graphene increased, the percent recovery only increased to 90%.
 - J. No; as the mass of added graphene increased, the percent recovery did not increase linearly.
13. In Experiment 2, how many of the solutions had a pH that is considered basic?
- A. 1
 - B. 3
 - C. 5
 - D. 6

**Passage III**

Two experiments examined the effect of 4 *growth inhibitors* (compounds that inhibit cell growth) on the growth of the fungus *Rhizopus nigricans*.

Experiment 1

Forty circular pieces of a certain type of bread, each 5.0 cm in diameter, were separated equally into 4 groups (Groups A–D). Each piece in Groups A–C was then sprayed with 1.0 mL of a 0.10% solution consisting of a mixture of water and 1 of 3 mold growth inhibitors: Group A with Inhibitor X, Group B with Inhibitor Y, and Group C with Inhibitor Z. Group D was sprayed with water only. Each piece was then placed into a separate petri dish, and each piece was treated with the same number of *R. nigricans* cells. The dishes, along with a beaker filled with water, were then sealed into a 32°C incubator.

Every 24 hr, for each of the next 5 days, the average *mold coverage* (the percentage of the surface area of the bread covered by *R. nigricans*) was determined for each group (see Table 1). During this period, the beaker was periodically refilled with water.

Table 1					
Group	Average mold coverage (percent) at:				
	24 hr	48 hr	72 hr	96 hr	120 hr
A	< 1	1	2	10	26
B	< 1	5	36	75	100
C	< 1	3	20	55	78
D	< 1	10	41	82	100

Experiment 2

Thirty more of the pieces of bread were separated equally into 3 groups (Groups E–G). These groups were then prepared in the same manner as Group A in Experiment 1, except that the inhibitor mixture sprayed on each piece of bread consisted of water and Inhibitor X only, and the concentration of Inhibitor X was different for each group: 0.01% (Group E), 0.05% (Group F), and 0.20% (Group G). Groups E–G were then incubated and analyzed as in Experiment 1 (see Table 2).

Table 2					
Group	Average mold coverage (percent) at:				
	24 hr	48 hr	72 hr	96 hr	120 hr
E	< 1	48	92	100	100
F	< 1	10	75	95	100
G	< 1	< 1	< 1	1	5

Tables adapted from D. Melnick, H. W. Vahlteich, and A. Hackett, "Sorbic Acid as a Fungistatic Agent for Foods XI. Effectiveness of Sorbic Acid in Protecting Cakes." ©1956 by the Institute of Food Technologists.

14. In the experiments, how many groups had an average mold coverage of 100% at 96 hr ?

F. 1
G. 2
H. 4
J. 7

15. What is the *minimum* number of days that were required to complete Experiment 2 ?

A. 3
B. 5
C. 15
D. 35

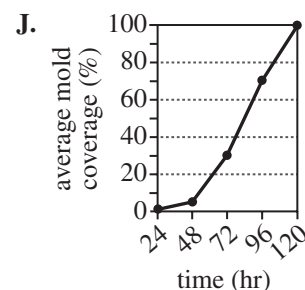
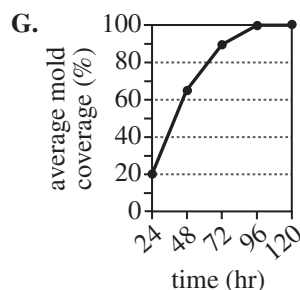
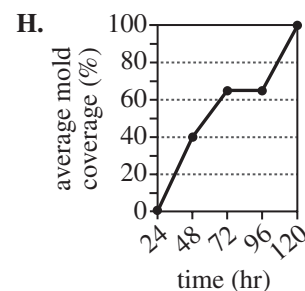
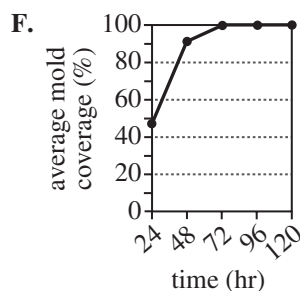
16. Which of the following conclusions about the change in average mold coverage is best supported by the results of Experiment 1 ? Over 120 hr of incubation, mold coverage:

F. decreased in Groups A and C, but increased in Groups B and D.
G. increased in Groups A and C, but decreased in Groups B and D.
H. decreased in all 4 groups.
J. increased in all 4 groups.

17. In Experiment 1, which group was most likely intended to serve as a control for the effect of inhibitors?

A. Group A
B. Group B
C. Group C
D. Group D

18. Suppose an Inhibitor X concentration of 0.08% had been tested in Experiment 2. Based on the results of Experiments 1 and 2, which of the following graphs shows plausible results for this concentration?





19. The biochemical reactions that provide the energy for the growth of *R. nigricans* occur in which organelle?

- A. Chloroplast
- B. Endoplasmic reticulum
- C. Golgi apparatus
- D. Mitochondrion

20. Which of the following expressions gives the average surface area of the bread, in square centimeters (cm^2), covered by *R. nigricans* for Group C at 72 hr in Experiment 1 ?

- F. $\pi \times (2.5 \text{ cm})^2 \times 20\%$
- G. $\pi \times (5.0 \text{ cm})^2 \times 20\%$
- H. $\pi \times (2.5 \text{ cm})^2 \times 78\%$
- J. $\pi \times (5.0 \text{ cm})^2 \times 78\%$



Passage IV

Introduction

A person is caught in a rainstorm. The person must travel westward along a straight path for a particular distance, L , in order to reach a shelter. Based on the number of raindrops falling on the person per second, N , and the total number of raindrops, S , that fall on the person while traveling the distance L , each of 4 students discusses whether the person should walk or run. The 4 students are given Figure 1 and the list of information that follows Figure 1.

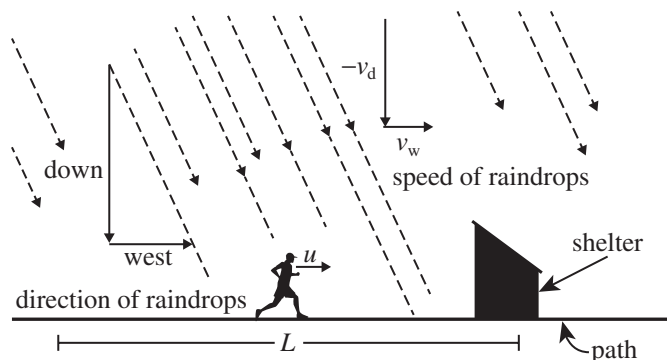


Figure 1

- The rain is falling downward and to the west. The *intensity* of the rainfall—the number of drops per second striking each square meter of ground—is constant.
- The velocity of any given raindrop has both a downward vertical component, $-v_{\text{down}}$ (or $-v_d$), and a westward horizontal component, v_{west} (or v_w). Both components are constant, and v_w is parallel to the path.
- The person's speed is represented by the symbol u .
- The time required for the person to travel the distance L is represented by the symbol t .

Student 1

The person should walk. Time t will be greater if the person walks than if the person runs, but N will be less, so S will be less.

Student 2

The person should run, because N will be less than if the person walks. Moreover, both t and S will be less than if the person walks because the person will spend less time in the rain.

Student 3

The person's speed should equal v_w . At that speed, both N and S will have their minimum values, so both N and S will be smaller than if the person moves at a different speed.

Student 4

S will be approximately the same whether the person walks or runs. N will be less if the person walks than if the person runs, but t will be greater. N will be greater if the person runs, but t will be less.

- Which student proposes that u should be greater, Student 1 or Student 2 ?
 - Student 1, because Student 1 proposes that the person should walk.
 - Student 1, because Student 1 proposes that the person should run.
 - Student 2, because Student 2 proposes that the person should walk.
 - Student 2, because Student 2 proposes that the person should run.
- Which of the following expressions involving the person's speed and a component of the falling raindrops' velocity is consistent with Student 3's conclusions?
 - $u = \frac{1}{v_w}$
 - $u = \frac{1}{v_d}$
 - $u = v_w$
 - $u = v_d$
- Based on the Introduction, the intensity of rainfall is best represented by which of the following expressions?
 - Intensity = number of drops per second per square meter
 - Intensity = square meters per number of drops per second
 - Intensity = total number of drops per square meter
 - Intensity = square meters per total number of drops



24. Suppose the person, having traveled westward the distance L , turns around and travels eastward along the path through the distance L . Where will the raindrops more likely strike the person, in the front or in the back?
- F. In the front, because the person will be traveling in the same direction as the rain's horizontal motion.
 - G. In the front, because the person will be moving in the direction opposite to the rain's horizontal motion.
 - H. In the back, because the person will be traveling in the same direction as the rain's horizontal motion.
 - J. In the back, because the person will be moving in the direction opposite to the rain's horizontal motion.
25. Based on the Introduction, t can be calculated using which of the following expressions?
- A. $t = \frac{u}{L}$
 - B. $t = \frac{L}{v_d}$
 - C. $t = \frac{L}{u}$
 - D. $t = \frac{v_d}{L}$
26. The statement "In general, as u increases, N will also increase" is consistent with the predictions made by which of the following pairs of students, if any?
- F. Students 1 and 2 only
 - G. Students 1 and 4 only
 - H. Students 2 and 3 only
 - J. The statement is not consistent with the predictions made by any of these pairs of students.
27. Student 4 would most likely agree with which of the following statements about N and t ?
- A. $N \times t$ is approximately constant.
 - B. $N \div t$ is approximately constant.
 - C. $N + t$ is approximately constant.
 - D. $N - t$ is approximately constant.

**Passage V**

Biodiesel is a fuel that is produced when oil from plant sources reacts with ethanol in the presence of a catalyst. In an experiment, students extracted oil from *spent coffee grounds* (SCGs, coffee grounds that have already been used to brew coffee) and then determined how much biodiesel was produced when the oil was reacted with ethanol and sulfuric acid (H_2SO_4).

Experiment

Two different types of coffee beans (Type A and Type R) were obtained. Each type was roasted, ground, and then used to brew coffee. The SCGs from each type were then dried. Three separate 50 g samples (S1–S3) were taken from each type of dried SCG. Then Steps 1–5 were performed with each sample:

1. To extract oil from the SCGs, the sample was mixed with 200 mL of heptane (a nonpolar solvent) in a flask and stirred for 60 min at 1 of 3 extraction temperatures: 22°C (S1), 70°C (S2), or 95°C (S3).
2. The contents of the flask were filtered. The liquid portion was placed in a beaker and then heated until all the heptane evaporated, leaving oil as a residue.
3. The mass of the extracted oil was determined.
4. Ethanol (5.0 g) and H_2SO_4 (1.5 g) were each added to the beaker. The contents were then stirred for 24 hr at 22°C.
5. The mass of biodiesel produced was determined, and, based on the mass of extracted oil, the percent yield of biodiesel was calculated.

The results are shown in the table.

Coffee bean type	Sample	Extraction temperature (°C)	Mass of extracted oil (g)	Percent yield of biodiesel
A	S1	22	3.5	99
	S2	70	4.3	95
	S3	95	4.7	100
R	S1	22	2.7	81
	S2	70	3.1	94
	S3	95	3.3	100

Table adapted from Sophie Bendall et al., "Showcasing Chemical Engineering Principles through the Production of Biodiesel from Spent Coffee Grounds." ©2015 by American Chemical Society and Division of Chemical Education, Inc.

28. For each of the types of SCGs, which of the following statements describes a difference between the oil extraction procedure for the S2 samples and the S3 samples? The S2 samples were extracted:

- F. at 70°C, whereas the S3 samples were extracted at 95°C.
- G. at 95°C, whereas the S3 samples were extracted at 70°C.
- H. using ethanol, whereas the S3 samples were extracted using heptane.
- J. using heptane, whereas the S3 samples were extracted using ethanol.

29. Suppose that another 50 g sample of Type R SCGs had been tested in the experiment and that the mass of oil extracted had been 2.6 g. The extraction temperature for this sample would most likely have been closest to which of the following temperatures?

- A. 15°C
- B. 50°C
- C. 85°C
- D. 110°C

30. Before the experiment, a student predicted that, for the Type R SCG samples, the higher the extraction temperature, the greater the percent yield of biodiesel. Do the results of the experiment support the student's prediction?

- F. Yes; as the extraction temperature increased, the percent yield of biodiesel increased only.
- G. No; as the extraction temperature increased, the percent yield of biodiesel decreased only.
- H. No; as the extraction temperature increased, the percent yield of biodiesel increased and then decreased.
- J. No; as the extraction temperature increased, the percent yield of biodiesel decreased and then increased.

31. Consider the Type A SCG sample for which the extraction temperature was the same as the temperature at which stirring occurred in Step 4. The mass of extracted oil from this sample was:

- A. 2.7 g.
- B. 3.3 g.
- C. 3.5 g.
- D. 4.7 g.



32. Consider the 2 samples that had the lowest percent yields of biodiesel. What was the difference in the mass of extracted oil between these samples?
- F. 0.2 g
 - G. 0.4 g
 - H. 0.8 g
 - J. 1.2 g
33. According to the results of the experiment, for the Type A samples, as the extraction temperature increased from 22°C to 70°C to 95°C, the percent yield of biodiesel:
- A. increased only.
 - B. decreased only.
 - C. increased and then decreased.
 - D. decreased and then increased.
34. Based on the experimental procedures, was the extracted oil from SCGs more likely polar or nonpolar?
- F. Polar; the oil was extracted using a polar solvent, and polar substances are typically soluble in polar solvents.
 - G. Polar; the oil was extracted using a nonpolar solvent, and polar substances are typically soluble in nonpolar solvents.
 - H. Nonpolar; the oil was extracted using a polar solvent, and nonpolar substances are typically soluble in polar solvents.
 - J. Nonpolar; the oil was extracted using a nonpolar solvent, and nonpolar substances are typically soluble in nonpolar solvents.



Passage VI

An mRNA strand contains nucleotides that determine the sequence of amino acids in a protein. Each nucleotide consists of a sugar, a phosphate group, and 1 of 4 nucleobases—adenine (A), cytosine (C), guanine (G), or uracil (U). A group of 3 nucleobases forms a codon, and each codon specifies either an amino acid or a stop.

Figure 1 is used to determine what each codon specifies. The first nucleobase is in the center circle, and the second and third nucleobases are in the second and third circles, respectively. The outer circle shows the specified amino acid or a stop. For example, UUG is 1 of 6 codons that specify leucine.

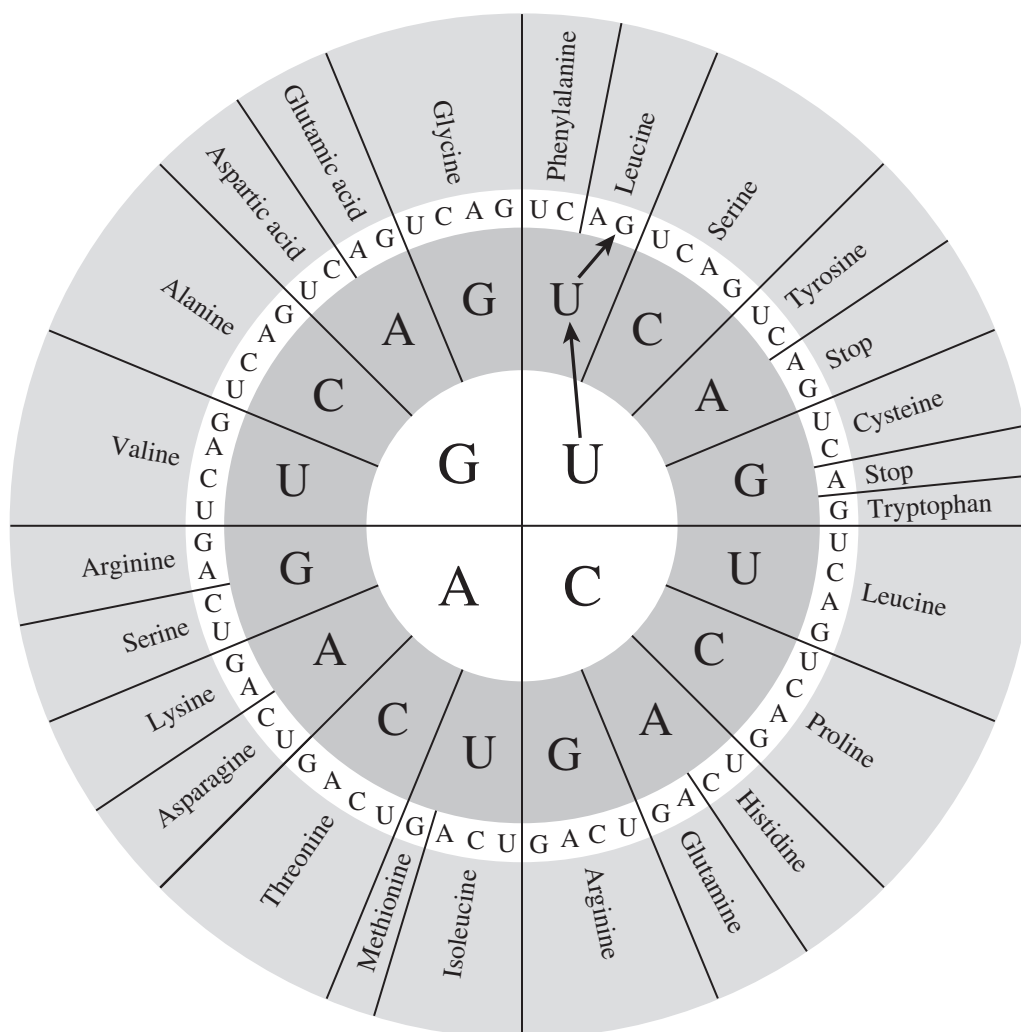


Figure 1



35. Which of the nucleobases can be the third nucleobase in a codon for isoleucine?
- A. A only
 - B. G only
 - C. A, C, and U only
 - D. A, C, G, and U
36. Suppose an mRNA strand contained 4 codons with the nucleobase sequence GCUAUGGCUAAG. What amino acid would be specified immediately after methionine in this strand?
- (Note: Assume the mRNA strand is read left to right.)
- F. Alanine
 - G. Glycine
 - H. Lysine
 - J. Tryptophan
37. In a complementary sequence, nucleobases are swapped such that A becomes U (and vice versa) and C becomes G (and vice versa). For example, the complementary sequence for GAA is CUU. Based on this information and Figure 1, which of the following sequences is a complementary sequence for asparagine?
- A. AAA
 - B. AAC
 - C. UUA
 - D. UUC
38. The codons that specify a stop share a common feature in that they all:
- F. begin with uracil.
 - G. end with adenine.
 - H. contain at least 1 guanine.
 - J. contain 3 different nucleobases.
39. Multiple codons that specify the same amino acid are said to demonstrate redundancy. Which of the following pairs of codons demonstrates redundancy?
- A. AAA and GGG
 - B. AGG and GUU
 - C. CUA and UUA
 - D. CUC and UCU
40. Consider the entire outer circle in Figure 1. Serine is specified by how many different codons?
- F. 2
 - G. 4
 - H. 6
 - J. 8

END OF TEST 4

STOP! DO NOT RETURN TO ANY OTHER TEST.

Scoring Guide

Raw Scores

The number of questions you answered correctly on each test section is a raw score. Because there are many forms of the ACT, each with different questions, the difficulty level varies between the forms. A raw score of 35 on one form of the mathematics test section, for example, may be about as difficult to earn as a raw score of 37 on another form of that test section.

Computing raw scores: To compute your raw scores, check your answers with the scoring information in the scoring keys and conversion table and do the following:

1. Mark a one (1) in the blank for each question answered correctly.
2. Count the number of correct answers for each of the four multiple-choice test sections.
3. Add up the total number correct for each category and test section and capture it as directed above its scoring key.

These numbers are your raw scores on the individual multiple-choice test sections. The highest raw score for a given test section is the number of questions included on that test section:

- English: 75
- Mathematics: 60
- Reading: 40
- Science: 40

Note: Raw scores are also provided as part of the Summary Report for each test section taken online in TestNav.

English Test 1 Section

English Scoring Key (for Form H31)

English Reporting Categories
(Capture raw scores/correct answers.)

Production of Writing (POW) = ____ of 23

Knowledge of Language (KLA) = ____ of 12

Conventions of Standard English (CSE) = ____ of 40

Total English (POW + KLA + CSE) = ____ of 75

English Number	Correct Answer	Correct (Mark 1)	Reporting Categories
1	C		CSE
2	J		KLA
3	A		POW
4	F		KLA
5	B		CSE
6	G		KLA
7	D		POW
8	J		POW
9	A		CSE
10	H		CSE
11	B		CSE
12	J		CSE
13	B		CSE
14	F		CSE
15	C		POW
16	H		KLA
17	B		POW
18	G		CSE
19	C		POW
20	J		CSE
21	D		CSE
22	F		POW
23	D		POW
24	J		CSE
25	C		POW

English Number	Correct Answer	Correct (Mark 1)	Reporting Categories
26	H		CSE
27	A		CSE
28	J		CSE
29	B		POW
30	G		POW
31	D		CSE
32	G		KLA
33	B		POW
34	F		CSE
35	B		POW
36	H		CSE
37	B		CSE
38	F		CSE
39	D		CSE
40	F		KLA
41	C		POW
42	H		CSE
43	D		CSE
44	J		CSE
45	C		POW
46	H		CSE
47	B		POW
48	F		POW
49	D		KLA
50	F		CSE

English Number	Correct Answer	Correct (Mark 1)	Reporting Categories
51	D		CSE
52	G		POW
53	C		CSE
54	F		POW
55	D		KLA
56	F		CSE
57	C		CSE
58	J		CSE
59	C		CSE
60	F		CSE
61	B		KLA
62	F		KLA
63	C		CSE
64	G		POW
65	B		CSE
66	G		POW
67	C		CSE
68	J		CSE
69	C		CSE
70	H		KLA
71	D		CSE
72	F		CSE
73	C		POW
74	F		KLA
75	D		POW

Mathematics Test 2 Section

Mathematics Scoring Key (for Form H31)

Mathematics Reporting Categories

(Capture raw scores/correct answers.)

Preparing for Higher Math (PHM) = ___ of 36

(A + F + G + N + S)

A = Algebra

F = Functions

G = Geometry

N = Number & Quantity

S = Statistics & Probability

Integrating Essential Skills (IES) = ___ of 24

Modeling (MDL) = ___ of 22

Total Mathematics (PHM + IES) = ___ of 60

Note: Do not include MDL in Total Mathematics raw score.

Math Number	Correct Answer	Correct (Mark 1)	Reporting Categories
1	B		IES
2	G		IES, MDL
3	C		IES, MDL
4	J		IES, MDL
5	B		IES, MDL
6	G		PHM-S, MDL
7	E		PHM-S, MDL
8	G		PHM-A
9	A		IES, MDL
10	F		IES, MDL
11	B		PHM-G
12	F		IES, MDL
13	E		PHM-N
14	K		PHM-G, MDL
15	A		IES
16	H		PHM-A
17	B		PHM-A
18	H		PHM-S, MDL
19	E		IES
20	G		PHM-N
21	C		IES

Math Number	Correct Answer	Correct (Mark 1)	Reporting Categories
22	H		PHM-F
23	C		PHM-S, MDL
24	G		PHM-G
25	D		PHM-F
26	K		IES, MDL
27	A		PHM-G
28	J		IES
29	B		IES
30	H		IES, MDL
31	C		PHM-S, MDL
32	H		PHM-F
33	C		PHM-G
34	K		PHM-A
35	D		IES, MDL
36	J		PHM-N
37	A		PHM-F
38	J		PHM-A, MDL
39	E		PHM-N
40	G		IES
41	C		IES
42	K		PHM-N
43	B		IES
44	F		PHM-A
45	A		IES, MDL
46	G		IES
47	B		PHM-N
48	K		PHM-G, MDL
49	C		PHM-A
50	J		IES
51	E		PHM-G
52	F		PHM-F
53	D		IES, MDL
54	K		PHM-S
55	B		PHM-F, MDL
56	H		PHM-G
57	E		PHM-A
58	G		PHM-F, MDL
59	B		PHM-F
60	J		IES

Reading Test 3 Section

Reading Scoring Key (for Form H31)

Reading Reporting Categories

(Capture raw scores/correct answers.)

Key Ideas & Details (KID) = ___ of 23

Craft & Structure (CS) = ___ of 10

Integration of Knowledge & Ideas (IKI) = ___ of 7

Total Reading (KID + CS + IKI) = ___ of 40

Reading Number	Correct Answer	Correct (Mark 1)	Reporting Categories
1	D		CS
2	H		KID
3	B		KID
4	F		KID
5	A		KID
6	H		CS
7	A		KID
8	F		KID
9	D		KID
10	G		KID
11	B		CS
12	H		CS
13	D		KID
14	J		KID

Reading Number	Correct Answer	Correct (Mark 1)	Reporting Categories
15	B		KID
16	G		KID
17	A		KID
18	H		IKI
19	B		IKI
20	H		IKI
21	D		CS
22	H		KID
23	B		KID
24	F		CS
25	B		CS
26	H		KID
27	A		IKI
28	J		KID
29	C		CS
30	F		CS
31	D		KID
32	J		KID
33	D		KID
34	H		IKI
35	A		KID
36	G		IKI
37	B		KID
38	F		CS
39	A		KID
40	F		IKI

Science Test 4 Section

Science Scoring Key (for Form H31)

Science Reporting Categories

(Capture raw scores/correct answers.)

Interpretation of Data (IOD) = ____ of 19

Scientific Investigation (SIN)= ____ of 10

Evaluation of Models, Inferences
& Experimental Results (EMI) = ____ of 11

Total Science (IOD + SIN + EMI) = ____ of 40

Science Number	Correct Answer	Correct (Mark 1)	Reporting Categories
1	C		IOD
2	J		IOD
3	D		IOD
4	G		SIN
5	B		EMI
6	G		EMI
7	B		SIN
8	F		SIN
9	C		IOD
10	G		SIN
11	C		SIN
12	J		EMI
13	A		IOD
14	F		IOD

Science Number	Correct Answer	Correct (Mark 1)	Reporting Categories
15	B		SIN
16	J		EMI
17	D		SIN
18	J		SIN
19	D		IOD
20	F		IOD
21	D		EMI
22	H		EMI
23	A		IOD
24	G		EMI
25	C		IOD
26	G		EMI
27	A		EMI
28	F		SIN
29	A		SIN
30	F		EMI
31	C		IOD
32	G		IOD
33	D		IOD
34	J		EMI
35	C		IOD
36	F		IOD
37	C		IOD
38	F		IOD
39	C		IOD
40	H		IOD

Scale Scores

To adjust for the small differences among different forms of the ACT test, raw scores are converted into scale scores. Scale scores appear on reports sent to your school.

When your raw scores are converted into scale scores, it becomes possible to compare your scores with those of examinees who took different test forms. For example, a scale score of 26 on the mathematics test section has the same meaning for any form of the ACT.

Converting Raw Scores to Scale Scores

Each ACT test section generates a single scale score of 1–36. Use the Conversion of Raw Scores to Scale Scores table to convert your raw scores to scale scores. For each test section:

- 1. Locate your raw score, or the range of raw scores that includes it, in the conversion table.
- 2. Locate the cell in the Scale Score column of the row that corresponds to that raw score to identify the scale score for that test section.
- 3. As you determine your scale scores, enter them in the blanks below.

Test section scale scores:

English = ____
Mathematics = ____
Reading = ____
Science = ____

Calculating a Composite Score

An ACT test generates a single Composite score of 1–36. Compute the Composite score by averaging the four scale scores. To do this:

- 1. Add your four scale scores. Enter this sum of scores in the blank below.
- 2. Divide the sum by 4. If the resulting number ends in a fraction, round it to the nearest whole number. (Round down any fraction less than one-half, except for averages lower than one; round up any fraction that is one-half or more and/or less than one.)
- 3. Enter this number in the blank below. This is your Composite score.

Composite of scale scores:

Sum of scores = ____
Composite score (sum ÷ 4) = ____

Note: If you left a test section completely blank and marked no items, do not list a scale score for that section and do not calculate a Composite score.

Conversion of Raw Scores to Scale Scores (for Form H31)

Scale Score	Raw Score Test 1: English	Raw Score Test 2: Mathematics	Raw Score Test 3: Reading	Raw Score Test 4: Science
36	74–75	59–60	38–40	40
35	71–73	57–58	37	39
34	69–70	55–56	36	38
33	68	54	35	37
32	67	52–53	34	—
31	66	51	33	36
30	65	49–50	32	35
29	64	47–48	—	34
28	62–63	46	31	33
27	61	43–45	30	32
26	59–60	40–42	—	31
25	56–58	38–39	29	29–30
24	53–55	36–37	27–28	27–28
23	50–52	34–35	26	25–26
22	48–49	33	24–25	24
21	44–47	32	23	22–23
20	41–43	30–31	22	21
19	39–40	28–29	21	19–20
18	37–38	26–27	19–20	17–18
17	35–36	23–25	18	15–16
16	33–34	19–22	17	14
15	30–32	15–18	16	13
14	27–29	12–14	14–15	11–12
13	25–26	10–11	13	10
12	23–24	8–9	11–12	9
11	20–22	7	9–10	8
10	17–19	6	8	7
9	14–16	5	7	6
8	12–13	4	6	5
7	10–11	3	5	4
6	8–9	—	4	3
5	6–7	2	3	—
4	5	—	—	2
3	3–4	1	2	1
2	2	—	1	—
1	0–1	0	0	0



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